



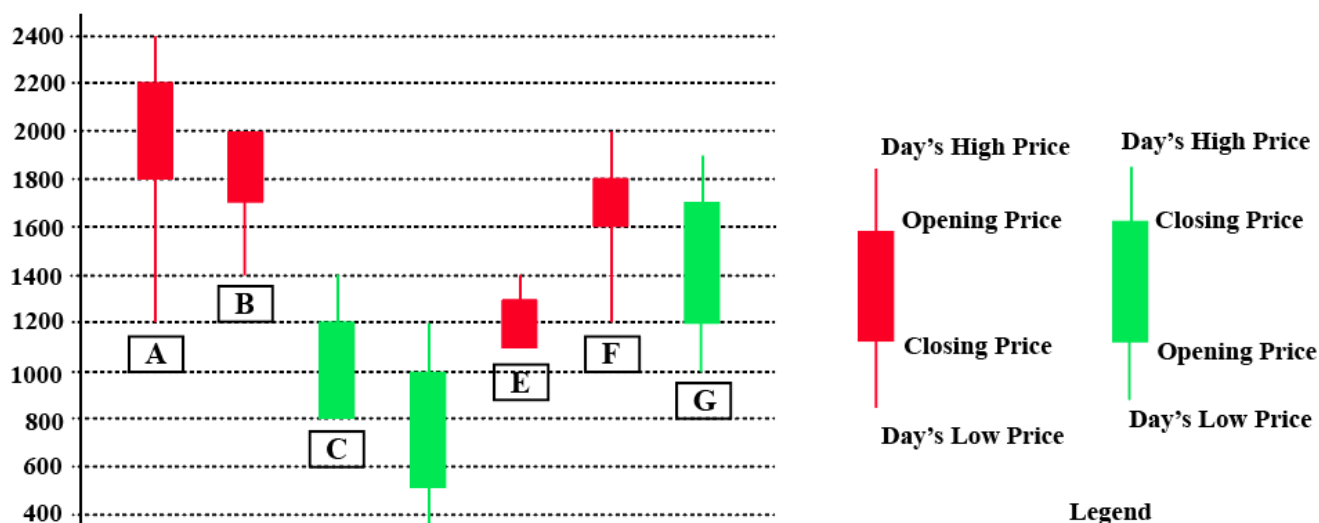
Top 54 CAT Special Charts Questions With Video Solutions

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Questions

Instructions

The chart below shows the price data for seven shares - A, B, C, D, E, F, and G as a candlestick plot for a particular day. The vertical axis shows the price of the share in rupees. A share whose closing price (price at the end of the day) is more than its opening price (price at the start of the day) is called a bullish share; otherwise, it is called a bearish share. All bullish and bearish shares are shown in green and red colour respectively.



Question 1

Daily loss for a share is defined as $(\text{Opening price} - \text{Closing price}) / (\text{Opening price})$. Which among the shares A, B, F and G had the highest daily loss on that day?

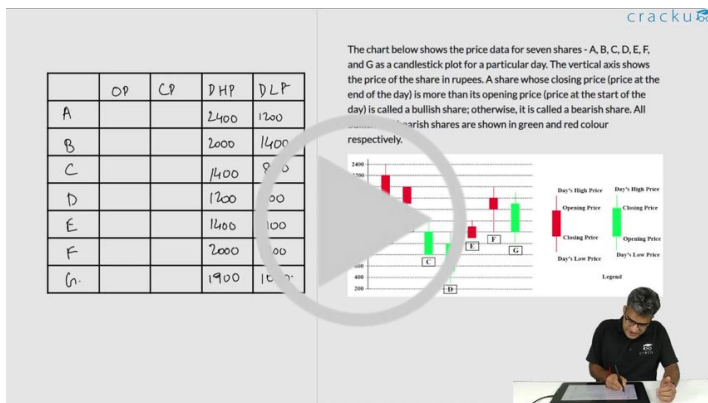
- A G
- B B
- C A
- D F

	OP	CP	DHP	DLP
A			2400	1200
B			2000	1400
C			1400	600
D			1200	200
E			1400	1000
F			2000	1200
G			1900	800

VIDEO SOLUTION

Question 2

Daily Share Price Variability (SPV) is defined as $(\text{Day's high price} - \text{Day's low price}) / (\text{Average of the opening and closing prices during the day})$. How many shares had an SPV greater than 0.5 on that day?

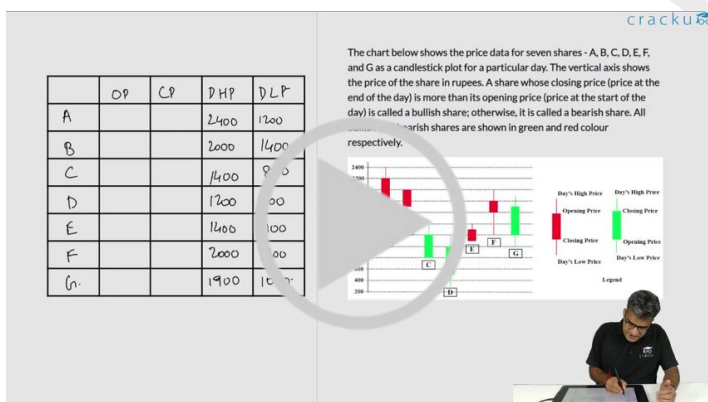


VIDEO SOLUTION

Question 3

Daily Share Price Variability (SPV) is defined as $(\text{Day's high price} - \text{Day's low price}) / (\text{Average of the opening and closing prices during the day})$. Which among the shares A, C, D and F had the highest SPV on that day?

- A F
- B A
- C D
- D C



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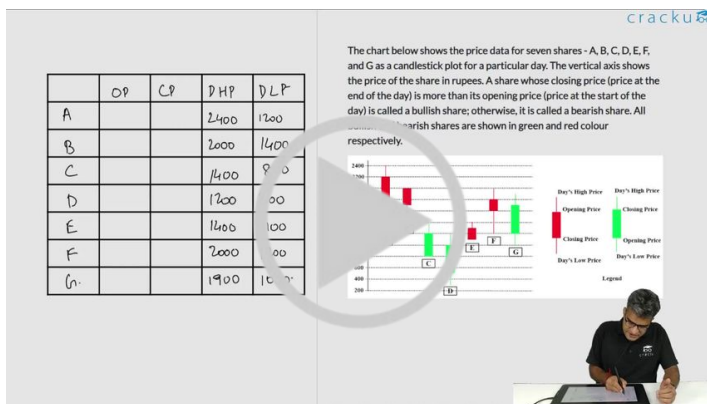
Question 4

What would have been the percentage wealth gain for a trader, who bought equal numbers of all bullish shares at opening price and sold them at their day's high?

- A 80%
- B 50%

C 72%

D 100%



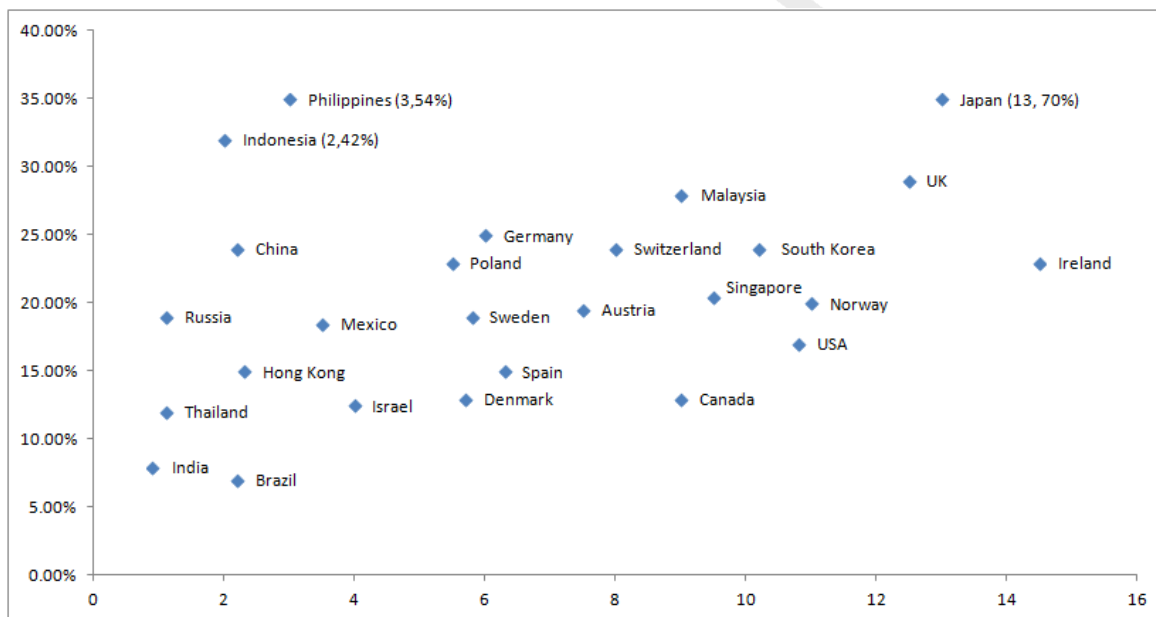
VIDEO SOLUTION

Instructions

Directions for the next three questions:

Answer the following questions based on the information given below: Telecom operators get revenue from transfer of data and voice. Average revenue received from transfer of each unit of data is known as ARDT. In the diagram below, the revenue received from data transfer as percentage of total revenue received and the ARDT in US Dollars (USD) are given for various countries.

The X axis represents the ARDT values and the Y axis represents the % of revenue from data transfer as a percentage of the total revenue.



Question 5

If the total revenue received is the same for the pairs of countries listed in the choices below, choose the pair that has approximately the same volume of data transfer.

- A Philippines and Austria
- B Canada and Poland
- C Germany and USA

- D UK and Spain
- E Denmark and Mexico

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Question 6

It was found that the volume of data transfer in India is the same as that of Singapore. Then which of the following statements is true?

- A Total revenue is the same in both countries.
- B Total revenue in India is about 2 times that of Singapore.
- C Total revenue in India is about 4 times that of Singapore.
- D Total revenue in Singapore is about 2 times that of India.
- E Total revenue in Singapore is about 4 time that of India.

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Question 7

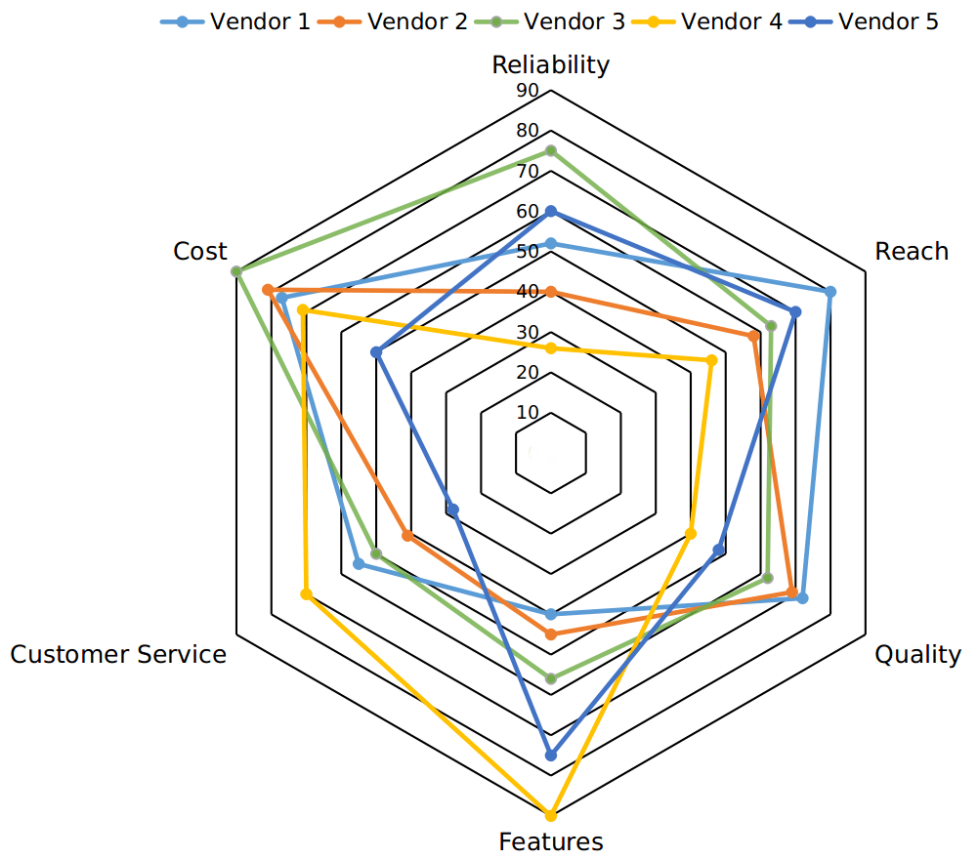
It is expected that by 2010, revenue from data transfer as a percentage of total revenue will triple for India and double for Sweden. Assume that in 2010, the total revenue in India is twice that of Sweden and that the volume of data transfer is the same in both the countries. What is the percentage increase of ARDT in India if there is no change in ARDT in Sweden?

- A 400%
- B 550%
- C 800%
- D 950%
- E cannot be determined

[VIEW SOLUTION](#)

Instructions

Five vendors are being considered for a service. The evaluation committee evaluated each vendor on six aspects - Cost, Customer Service, Features, Quality, Reach, and Reliability. Each of these evaluations are on a scale of 0 (worst) to 100 (perfect). The evaluation scores on these aspects are shown in the radar chart. For example, Vendor 1 obtains a score of 52 on Reliability, Vendor 2 obtains a score of 45 on Features and Vendor 3 obtains a score of 90 on Cost.



Question 8

List of all the vendors who are among the top two scorers on the maximum number of aspects is:

- A Vendor 2, Vendor 3 and Vendor 4
- B Vendor 1 and Vendor 5
- C Vendor 2 and Vendor 5
- D Vendor 1 and Vendor 2

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Five vendors are being considered for a service. The evaluation committee evaluated each vendor on six aspects - Cost, Customer Service, Features, Quality, Reach, and Reliability. Each of these evaluations are on a scale of 0 (worst) to 100 (perfect). The evaluation scores on these aspects are shown in the radar chart. For example, Vendor 1 obtains a score of 52 on Reliability, Vendor 2 obtains a score of 45 on Features and Vendor 3 obtains a score of 90 on Cost.

Question (1)
On which aspect is the median score of the five vendors the least?

Reliability - 52

VIDEO SOLUTION

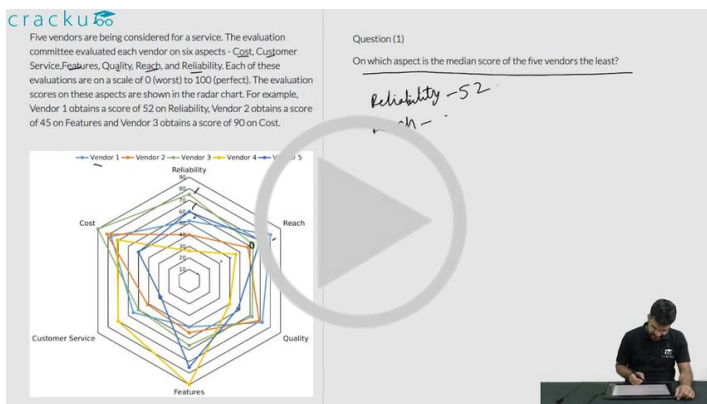
Question 9

A vendor's final score is the average of their scores on all six aspects. Which vendor has the highest final score?

- A Vendor 4
- B Vendor 2

C Vendor 1

D Vendor 3



VIDEO SOLUTION



Question 10

On which aspect is the median score of the five vendors the least?

- A Customer Service
- B Cost
- C Reliability
- D Quality



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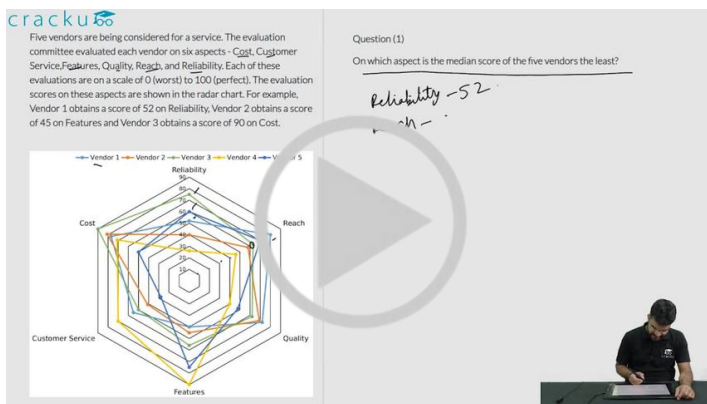
Question 11

List of all the vendors who are among the top three vendors on all six aspects is:

- A Vendor 1 and Vendor 3
- B None of the Vendors

C Vendor 3

D Vendor 1



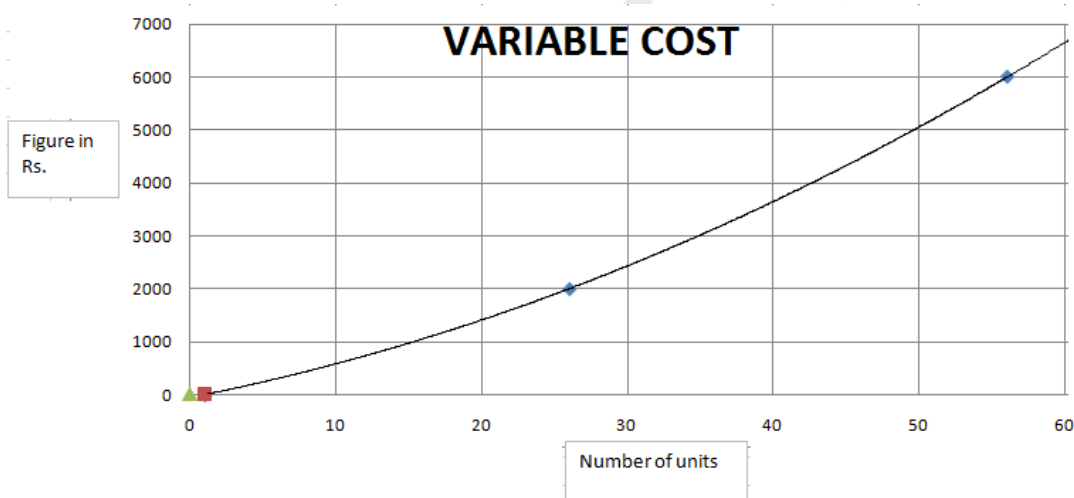
VIDEO SOLUTION

Instructions

Answer these questions based on the following Information:

ABC Ltd. produces widgets for which the demand is unlimited and they can sell all of their production. The graph below describes the monthly variable costs incurred by the company as a function of the quantity produced. In addition, operating the plant for one shift results in a fixed monthly cost of Rs. 800. Fixed monthly costs for second shift operation are estimated at Rs. 1200. Each shift operation provides capacity for producing 30 widgets per month.

Note: Average unit cost, $AC = \text{Total monthly costs} / \text{monthly production}$, and Marginal cost MC is the rate of change in total cost for unit change in quantity produced.



Question 12

Suppose that each widget sells for Rs. 150. What is the profit earned by ABC Ltd. in July, if it is known that 40 widgets were produced in this month? (Profit is defined as the excess of sales revenue over total cost).

- A 2400
- B 1600
- C 400
- D 0

[VIEW SOLUTION](#)



CAT Syllabus PDF



Question 13

Total production in July is 40 units. What is the approximate average unit cost for July?

- A 3600
- B 90
- C 140
- D 115

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Question 14

For monthly production level in the range of 30 to 60 Units,

- A AC is always higher than MC
- B AC is always lower than MC
- C AC is lower than MC up to a certain level and then is higher than MC
- D none of the above is true

[VIEW SOLUTION](#)

Question 15

Assume that the unit price is Rs. 150 and profit is defined as the excess of sales revenue over total costs. What is the monthly production level of ABC Ltd. at which the profit is highest?

- A 30
- B 50
- C 60
- D 40

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CAT Formula PDF



Question 16

From the data provided it can be inferred that, for production levels in the range of 0 to 60 units,

- A MC is an increasing function of production quantity.
- B MC is a decreasing function of production quantity.
- C initially MC is a decreasing function of production quantity, attains a minimum and then it is an increasing function of production quantity.
- D none of the above.

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Question 17

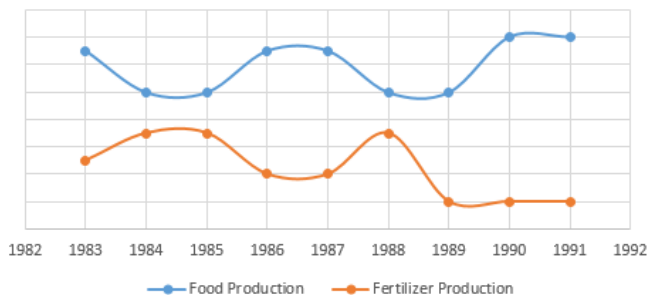
ABC Ltd. is considering increasing the production level. What is the approximate marginal cost of increasing production from its July level of 40 units.

- A 110
- B 130
- C 150
- D 160

[VIEW SOLUTION](#)

Instructions

Study the following graph and answer questions that follow. The x-axis denotes the years from 1983 to 1991



Question 18

Going according to previous trends, one can say that the Fertilizer Production has shown an anomalous behavior in which year?

- A 1985
- B 1984
- C 1991
- D 1989

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Question 19

From its apparent behaviour, the food production in year 1992 can be expected to .

- A go up
- B go down
- C remain the same as previous year.
- D nothing can be said.

[VIEW SOLUTION](#)

Question 20

If in 1988, the sum of the food and fertilizer production was 170 million tonnes, the value of food production must have been (approximately, in million tonnes) .

- A 90
- B 70
- C 100
- D Insufficient data

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Question 21

The sum of food and fertilizer production has shown a constant value for how many years?

- A None of the years
- B 2
- C 4
- D 5

[VIEW SOLUTION](#)

CAT Percentile Predictor



Question 22

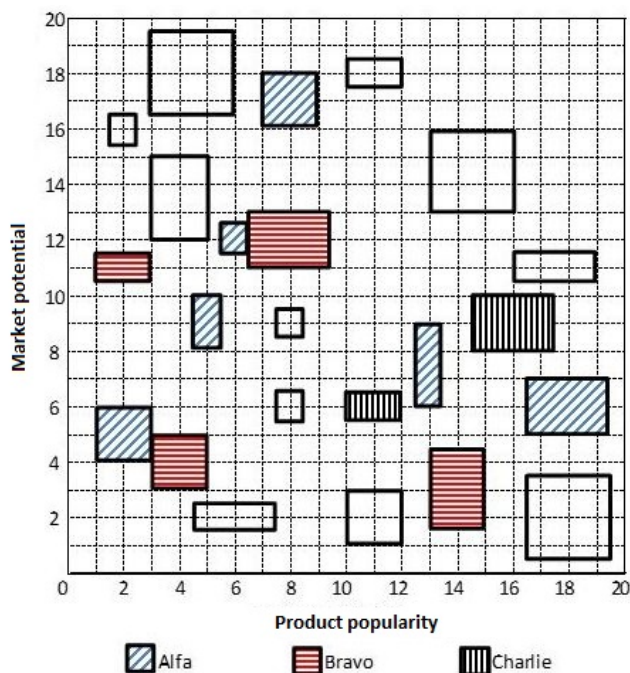
A scholar observed that if the production of fertilizers in 1989 had been the same as that in 1988, then the total fertilizer production for all the given years would have been 450 million tonnes. Using this information, and knowing that the food production has been plotted on the same scale, one may say that the food production in 1983 was (approximately, in million tonnes) .

- A 80
- B 130
- C 105
- D Cannot be determined

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Instructions

Each of the 23 boxes in the picture below represents a product manufactured by one of the following three companies: Alfa, Bravo and Charlie. The area of a box is proportional to the revenue from the corresponding product, while its centre represents the Product popularity and Market potential scores of the product (out of 20). The shadings of some of the boxes have got erased.



The companies classified their products into four categories based on a combination of scores (out of 20) on the two parameters - Product popularity and Market potential as given below:

	Promising	Blockbuster	Doubtful	No-hope
Product popularity score	>10	>10	≤ 10	≤ 10
Market potential score	>10	≤ 10	>10	≤ 10

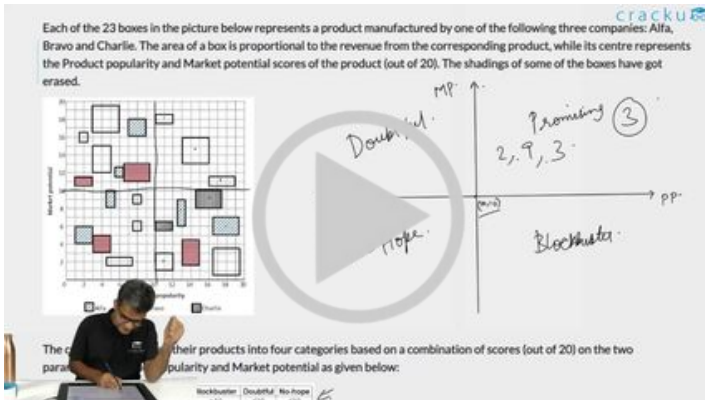
The following facts are known:

- Alfa and Bravo had the same number of products in the Blockbuster category.
- Charlie had more products than Bravo but fewer products than Alfa in the No-hope category.
- Each company had an equal number of products in the Promising category.
- Charlie did not have any product in the Doubtful category, while Alfa had one product more than Bravo in this category
- Bravo had a higher revenue than Alfa from products in the Doubtful category.
- Charlie had a higher revenue than Bravo from products in the Blockbuster category.
- Bravo and Charlie had the same revenue from products in the No-hope category.
- Alfa and Charlie had the same total revenue considering all products.

Question 23

Which of the following statements is NOT correct?

- A Alfa's revenue from Blockbuster products was the same as Charlie's revenue from Promising products
- B Bravo's revenue from Blockbuster products was greater than Alfa's revenue from Doubtful products
- C Bravo and Charlie had the same revenues from No-hope products
- D The total revenue from No-hope products was less than the total revenue from Doubtful products

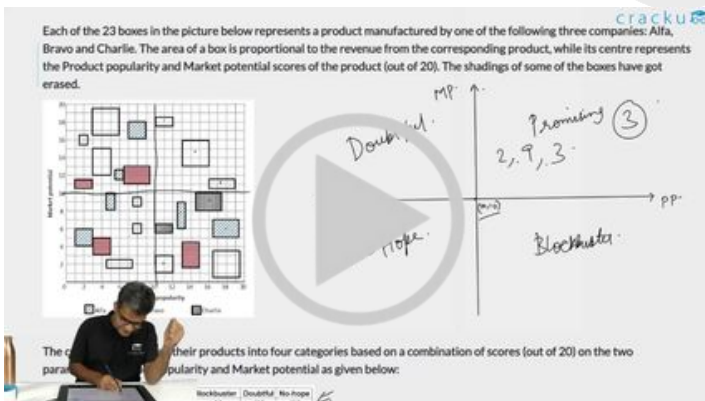


VIDEO SOLUTION

Question 24

Considering all companies' products, which product category had the highest revenue?

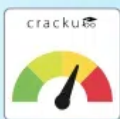
- A No-hope
- B Blockbuster
- C Doubtful
- D Promising



VIDEO SOLUTION



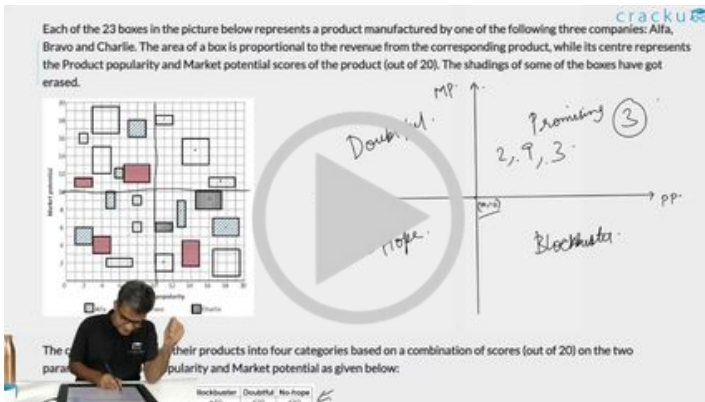
CAT Score Calculator



Question 25

Which of the following is the correct sequence of numbers of products Bravo had in No-hope, Doubtful, Promising and Blockbuster categories respectively?

- A 1,3,1,2
- B 1,3,1,3
- C 3,3,1,2
- D 2,3,1,2

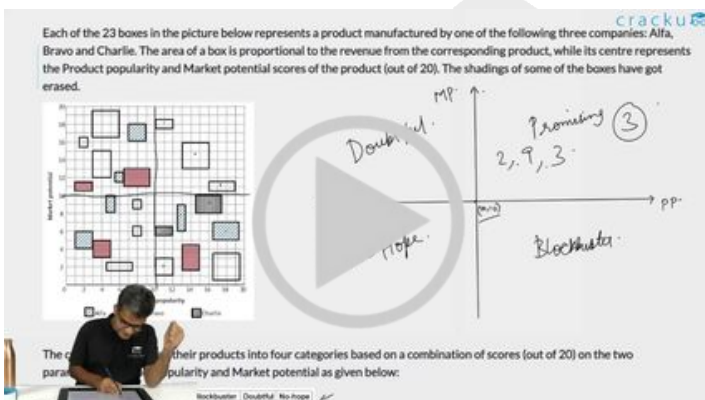


[VIDEO SOLUTION](#)

Question 26

If the smallest box on the grid is equivalent to revenue of Rs.1 crore, then what approximately was the total revenue of Bravo in Rs. crore?

- A 40
- B 24
- C 34
- D 30



[VIDEO SOLUTION](#)

Instructions

The table below gives information about four different crops, their different quality, categories and the regions where they are cultivated. Based on the information given in the table answer the questions below.

Type of Crop	Quality	Region
Crop - 1	High	R1, R2, R3, R4, R5
	Medium	R6, R7, R8
	Low	R9, R10, R11
Crop - 2	High	R5, R8, R12
	Medium	R9, R13
	Low	R6, R7, R8
Crop - 3	High	R2, R6, R7, R13
	Medium	R3, R9, R11
	Low	R1, R4
Crop - 4	High	R3, R10, R11
	Medium	R1, R2, R4
	Low	R5, R9

Question 27

How many low quality Crop-1 producing regions are either high quality Crop-4 producing regions or medium quality Crop-3 producing regions?

- A One
- B Two
- C Three
- D Zero

[VIEW SOLUTION](#)



Question 28

Which of the following statements is true?

- A All medium quality Crop-2 producing regions are also high quality Crop-3 producing regions.
- B All high quality Crop-1 producing regions are also medium and low Crop-4 producing regions.
- C There are exactly four Crop-3 producing regions, which also produce Crop-4 but not Crop-2.
- D Some Crop-3 producing regions produce Crop-1, but not high quality Crop-2.

[VIEW SOLUTION](#)

Question 29

How many regions produce medium qualities of Crop-1 or Crop-2 and also produce low quality of Crop-3 or Crop-4?

- A Zero
- B One
- C Two
- D Three

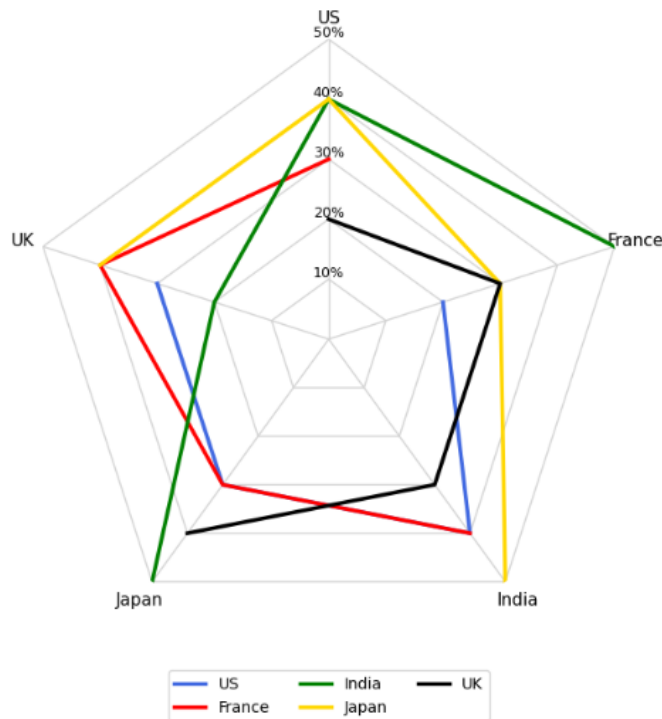
[VIEW SOLUTION](#)

Instructions

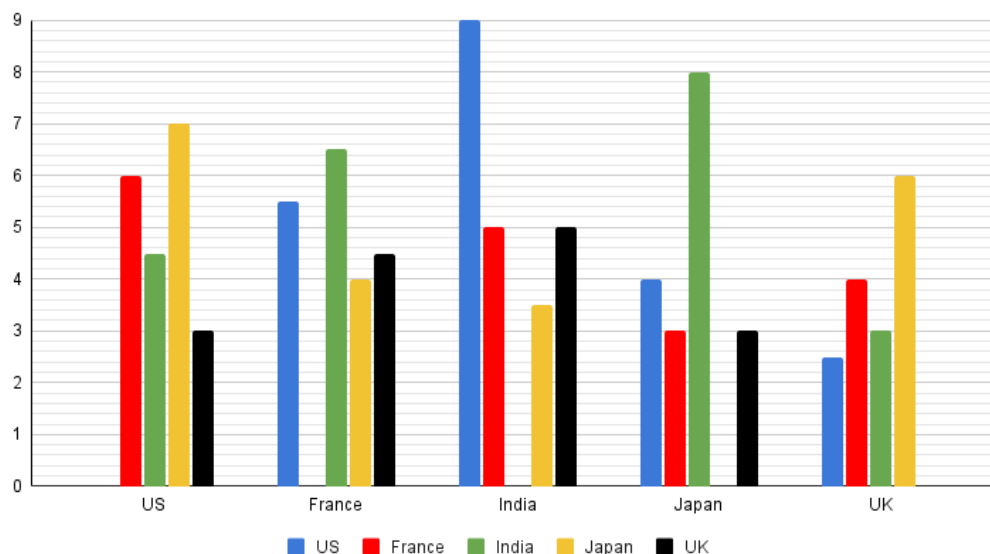
Five countries engage in trade with each other. Each country levies import tariffs on the other countries. The import tariff levied by Country X on Country Y is calculated by multiplying the corresponding tariff percentage with the total imports of Country X from Country Y.

The radar chart below depicts different import tariff percentages charged by each of the five countries on the others. For example, US (the blue line in the chart) charges 20%, 40%, 30%, and 30% import tariff percentages on imports from France, India, Japan, and UK, respectively. The bar chart depicts the import tariffs levied by each country on other countries. For example, US charged import tariff of 3 billion USD on UK.

Import Tariff Percentage Charged by Each Country from Other Countries



Import tariff in Billion USD charged by each country on other countries



Assume that imports from one country to another equals the exports from the latter to the former.

The trade surplus of Country X with Country Y is defined as follows.

Trade surplus = Exports from Country X to Country Y - Imports to Country X from Country Y.

A negative trade surplus is called trade deficit.

Question 30

How much is Japan's export to India worth?

- A 8.5 Billion USD
- B 16.0 Billion USD
- C 7.0 Billion USD
- D 1.75 Billion USD

[VIEW SOLUTION](#)



Question 31

Which among the following is the highest?

- A Exports by Japan to UK
- B Imports by US from France
- C Exports by France to Japan
- D Imports by France from India

[VIEW SOLUTION](#)

Question 32

Among France and UK, who has/have trade surplus(es) with US?

- A Neither France nor UK
- B Both France and UK
- C Only UK
- D Only France

[VIEW SOLUTION](#)

Question 33

What is the trade surplus/trade deficit of India with UK?

- A Surplus of 15.0 Billion USD
- B Deficit of 15.0 Billion USD
- C Surplus of 10.0 Billion USD
- D Deficit of 10.0 Billion USD

[VIEW SOLUTION](#)



CAT Daily Targets

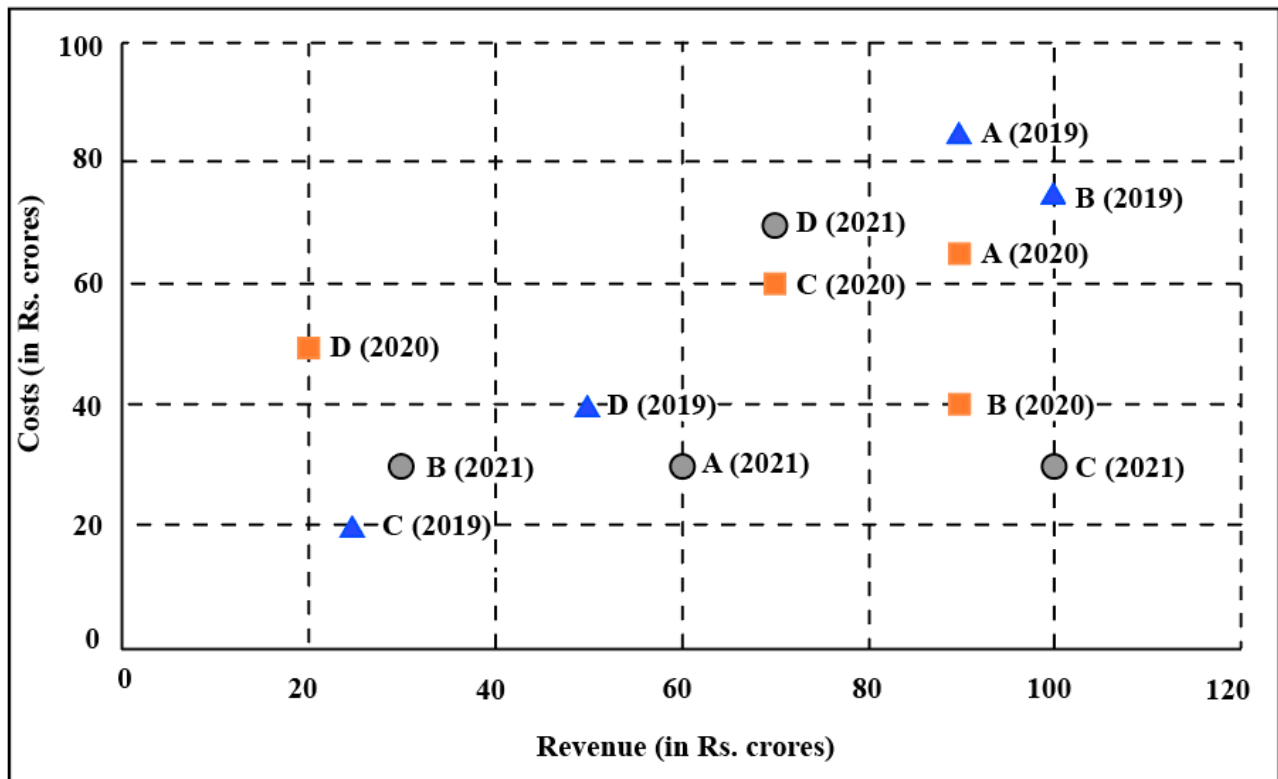
Step by step, heading towards 99+ percentile



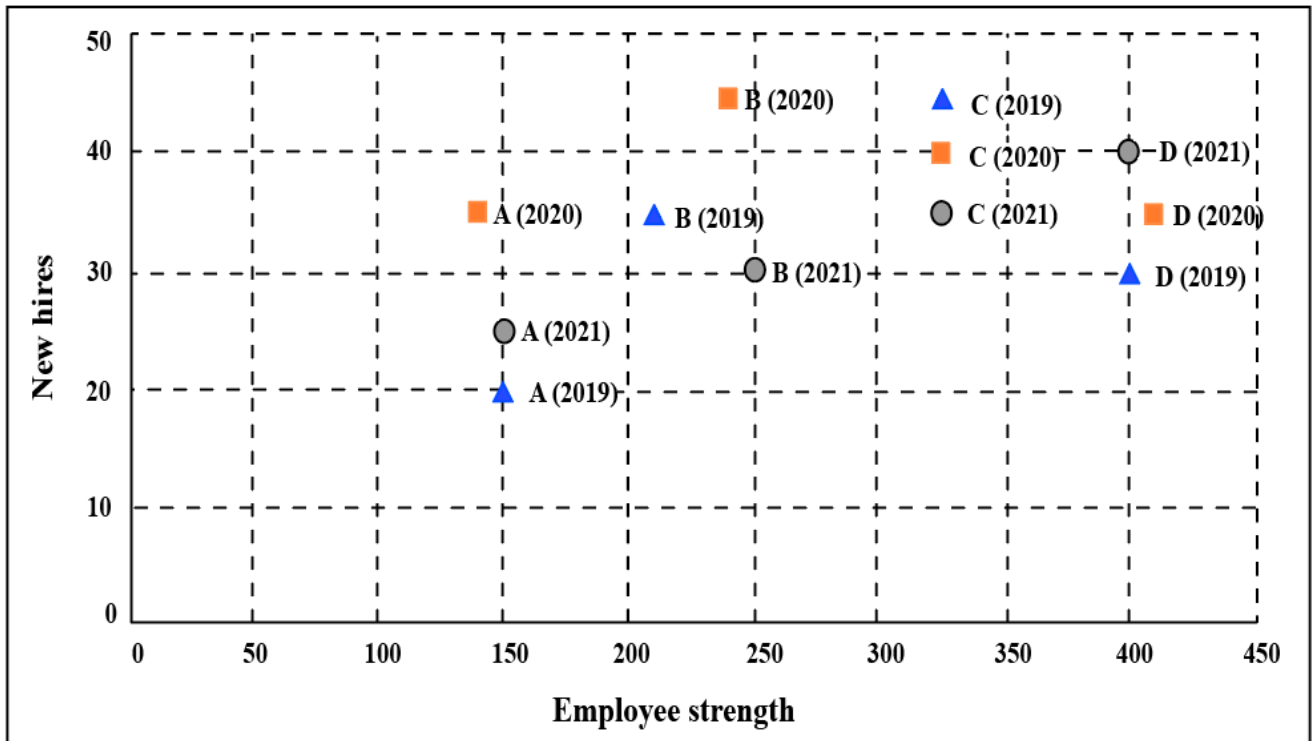
Instructions

The two plots below show data for four companies code-named A, B, C, and D over three years - 2019, 2020, and 2021.

The first plot shows the revenues and costs incurred by the companies during these years. For example, in 2021, company C earned Rs.100 crores in revenue and spent Rs.30 crores. The profit of a company is defined as its revenue minus its costs



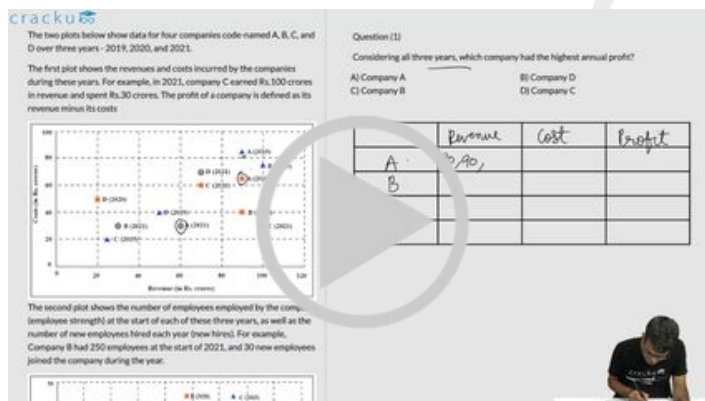
The second plot shows the number of employees employed by the company (employee strength) at the start of each of these three years, as well as the number of new employees hired each year (new hires). For example, Company B had 250 employees at the start of 2021, and 30 new employees joined the company during the year.



Question 34

Considering all three years, which company had the highest annual profit?

- A Company A
- B Company D
- C Company B
- D Company C



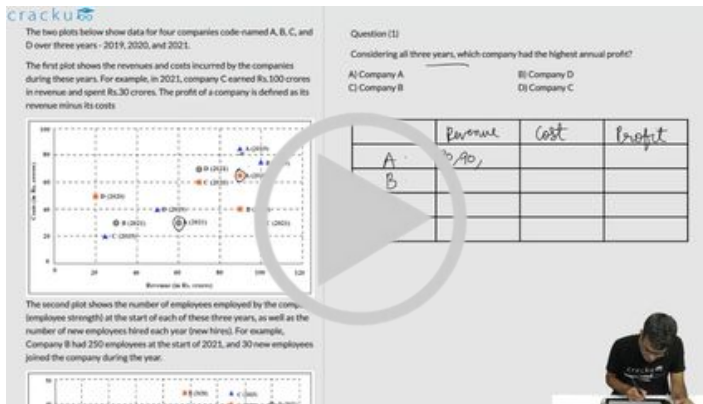
VIDEO SOLUTION

Question 35

Which of the four companies experienced the highest annual loss in any of the years?

- A Company C
- B Company A
- C Company B

D Company D

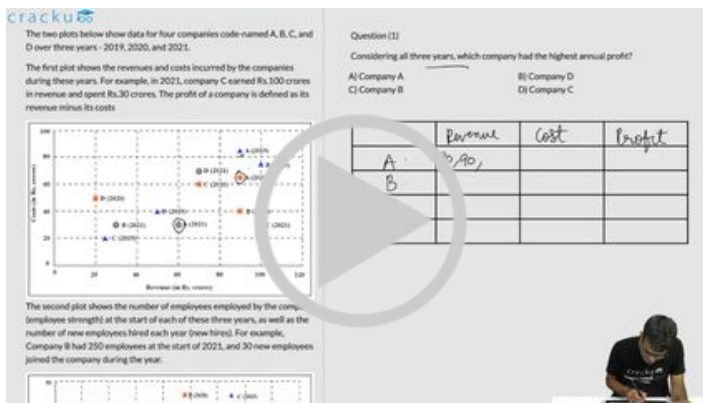


VIDEO SOLUTION

Question 36

The ratio of a company's annual profit to its annual costs is a measure of its performance. Which of the four companies had the lowest value of this ratio in 2019?

- A Company A
- B Company D
- C Company B
- D Company C



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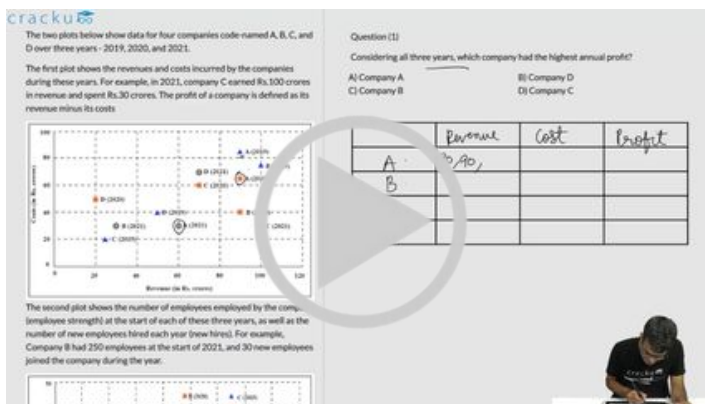
Question 37

The total number of employees lost in 2019 and 2020 was the least for:

- A Company B
- B Company D

C Company A

D Company C



VIDEO SOLUTION

Question 38

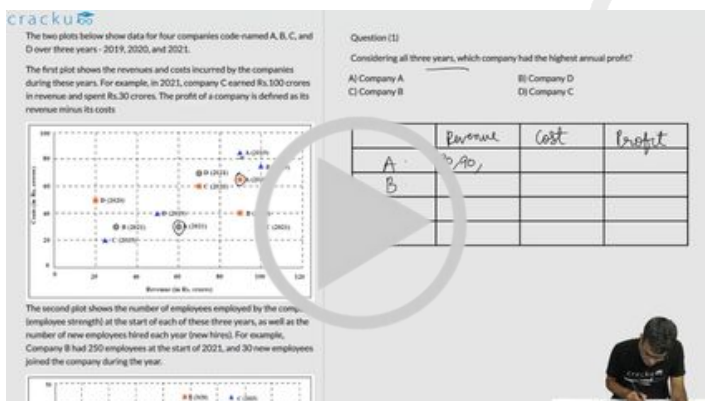
Profit per employee is the ratio of a company's profit to its employee strength. For this purpose, the employee strength in a year is the average of the employee strength at the beginning of that year and the beginning of the next year. In 2020, which of the four companies had the highest profit per employee?

A Company D

B Company C

C Company B

D Company A



VIDEO SOLUTION

Instructions

The two plots below give the following information about six firms A, B, C, D, E, and F for 2019 and 2023.

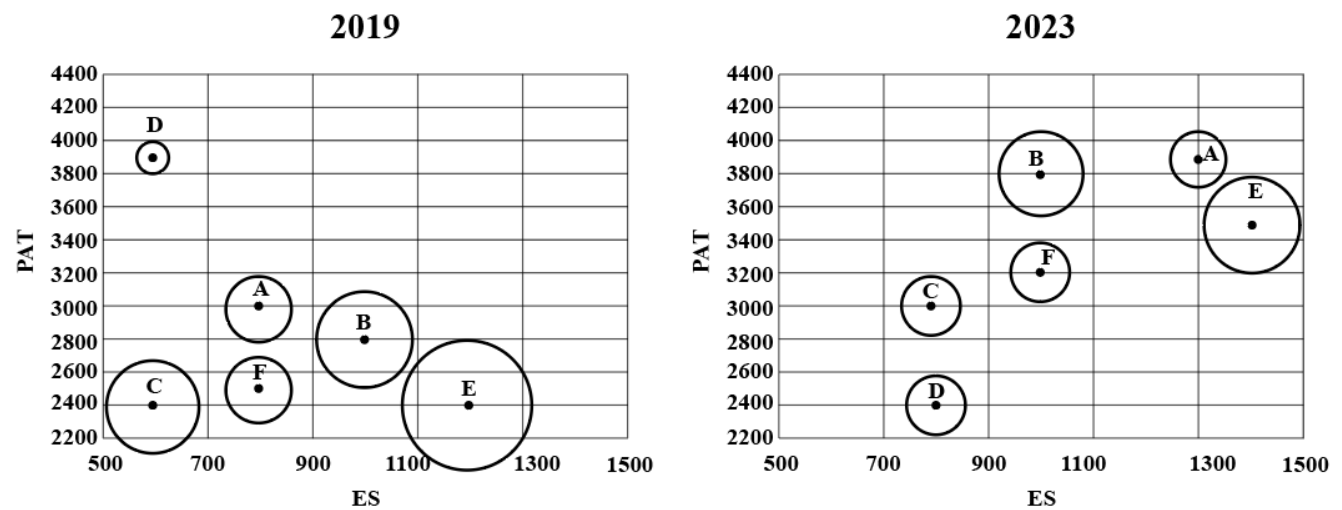
PAT: The firm's profits after taxes in Rs. crores,

ES: The firm's employee strength, that is the number of employees in the firm, and

PRD: The percentage of the firm's PAT that they spend on Research and Development (R&D).

In the plots, the horizontal and vertical coordinates of point representing each firm gives their ES and PAT values respectively. The PRD values of each firm are proportional to the areas around the points representing each firm. The areas are comparable between the two plots, i.e., equal areas in the two plots represent the

same PRD values for the two years.



Question 39

Which among the firms A, C, E, and F had the maximum PAT per employee in 2023?

- A Firm E
- B Firm A
- C Firm F
- D Firm C

The two plots below give the following information about six firms A, B, C, D, E, and F for 2019 and 2023.

PAT: The firm's profits after taxes in Rs. crores.

ES: The firm's employee strength, that is the number of employees in the firm.

PRD: The percentage of the firm's PAT that they spend on Research and Development (R&D).

In the plots, the horizontal and vertical coordinates of point representing a firm gives their ES and PAT values respectively. The PRD values of the firms are proportional to the areas around the points representing the firms. The PRD values of the firms are comparable between the two plots, i.e., equal PRD values in the two plots represent the same PRD values for the two years.

VIDEO SOLUTION

CAT Study Material

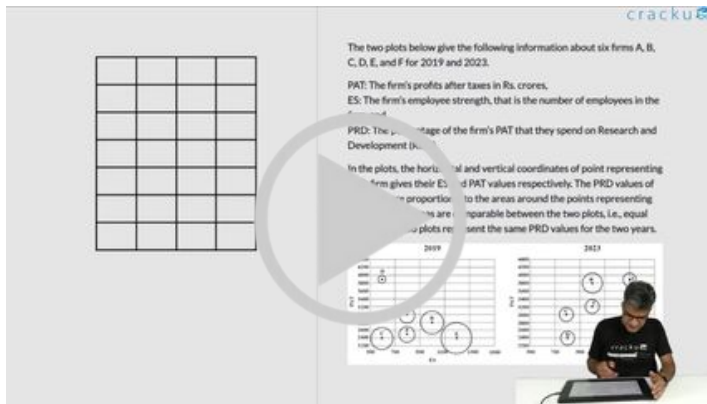
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Question 40

Assume that the annual rate of growth in PAT over the previous year (ARG) remained constant over the years for each of the six firms. Which among the firms A, B, C, and E had the highest ARG?

- A Firm A
- B Firm C
- C Firm E

D Firm B

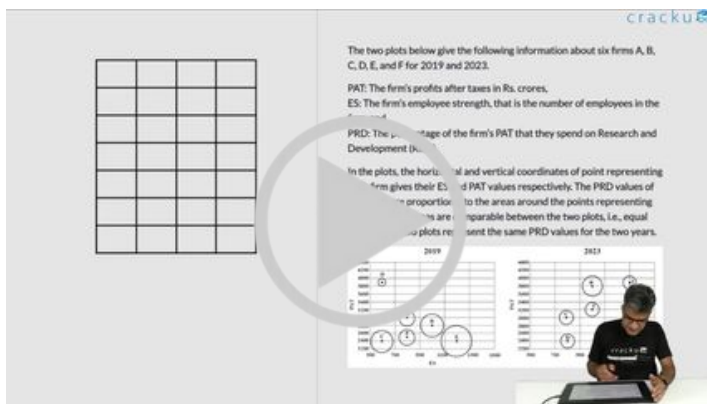


VIDEO SOLUTION

Question 41

The ratio of the amount of money spent by Firm C on R&D in 2019 to that in 2023 is closest to

- A 9 : 4
- B 5 : 6
- C 5 : 9
- D 9 : 5

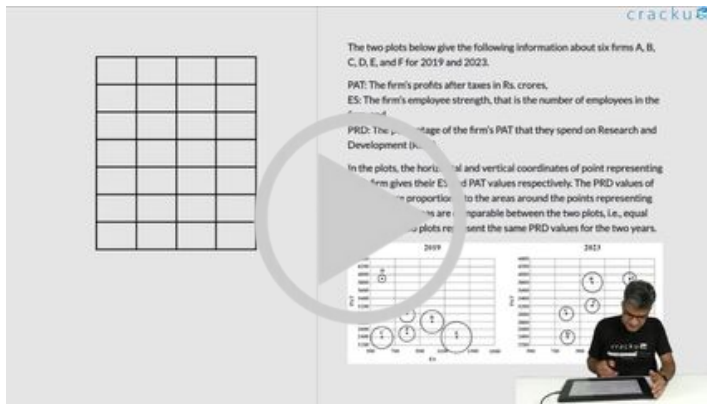


VIDEO SOLUTION

Question 42

Which among the firms C, D, E, and F had the least amount of R&D spending per employee in 2023?

- A Firm F
- B Firm D
- C Firm C
- D Firm E



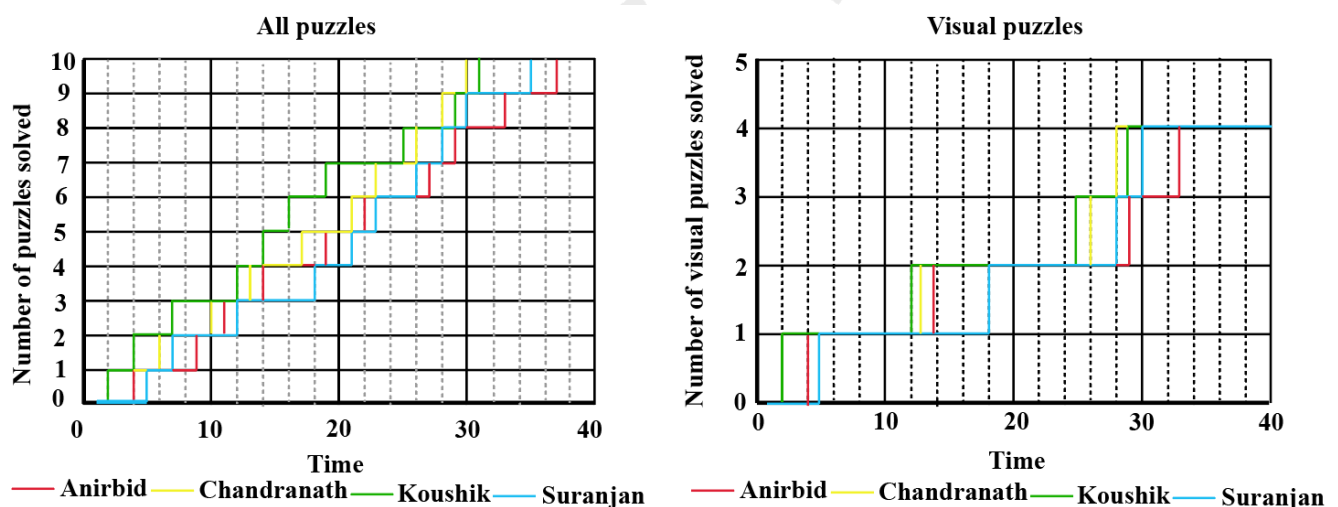
VIDEO SOLUTION



Instructions

Anirbid, Chandranath, Koushik, and Suranjan participated in a puzzle solving competition. The competition comprised 10 puzzles that had to be solved in the same sequence, i.e., a competitor got access to a puzzle as soon as they solved the previous puzzle. Some of the puzzles were visual puzzles and the others were number-based puzzles. The winner of the competition was the one who solved all puzzles in the least time.

The following charts describe their progress in the competition. The chart on the left shows the number of puzzles solved by each competitor at a given time (in minutes) after the start of the competition. The chart on the right shows the number of visual puzzles solved by each competitor at a given time (in minutes) after the start of the competition.



Question 43

At what number in the sequence was the fourth number-based puzzle?

VIEW SOLUTION

Question 44

Who had solved the largest number of puzzles by the 20-th minute from the start of the competition?

A Chandranath

- B Koushik
- C Suranjan
- D Anirbid

[VIEW SOLUTION](#)

Question 45

Which of the following is the closest to the average time taken by Anirbid to solve the number-based puzzles in the competition?

- A 3.3 minutes
- B 2.5 minutes
- C 3.8 minutes
- D 4.0 minutes

[VIEW SOLUTION](#)

100+ CAT Quant Questions



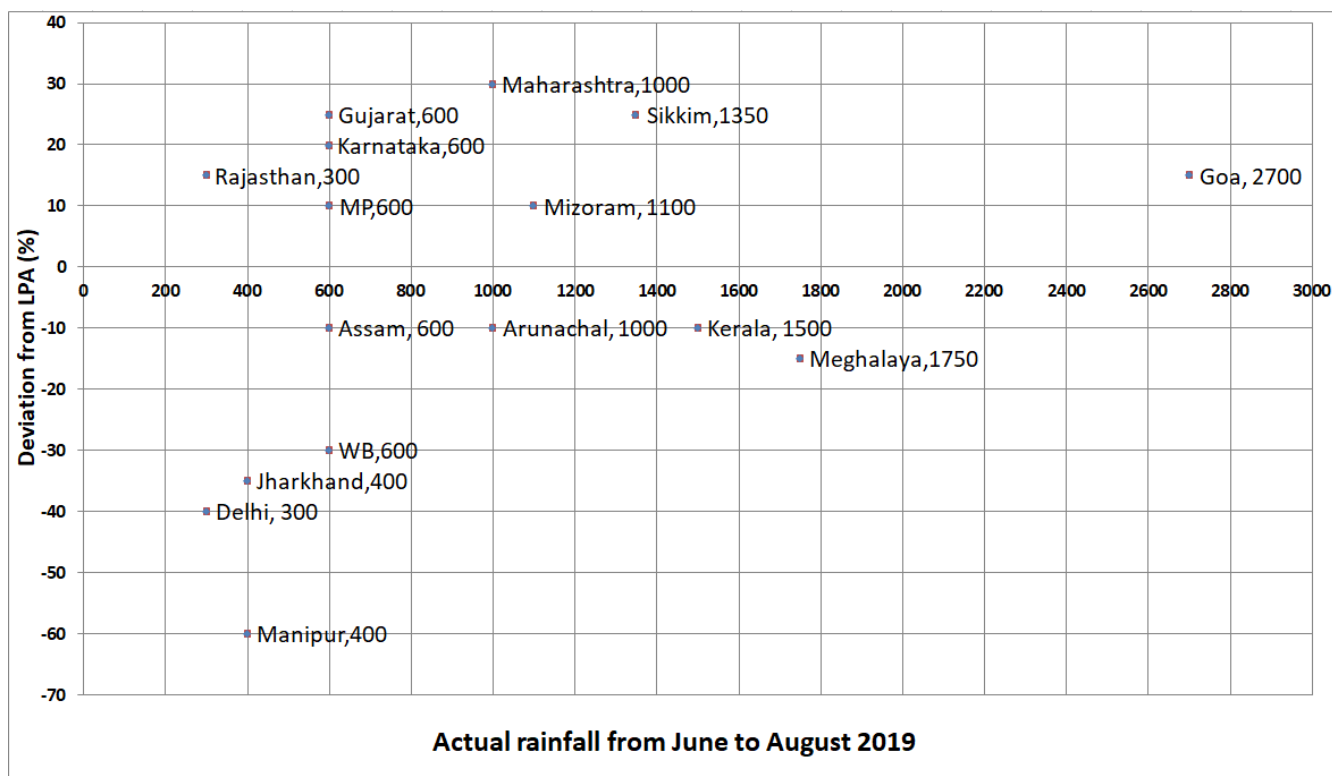
Question 46

How many minutes did Suranjan take to solve the third visual puzzle in the competition?

[VIEW SOLUTION](#)

Instructions

To compare the rainfall data, India Meteorological Department (IMD) calculated the Long Period Average (LPA) of rainfall during period June-August for each of the 16 states. The figure given below shows the actual rainfall (measured in mm) during June-August, 2019 and the percentage deviations from LPA of respective states in 2018. Each state along with its actual rainfall is presented in the figure.



Question 47

If a 'Low Monsoon State' is defined as a state with actual rainfall from June-August, 2019 of 750 mm or less, then what is the median 'deviation from LPA' (as defined in the Y-axis of the figure) of 'Low Monsoon States'?

- A -10%
- B 10%
- C -20%
- D -30%

cracku

To compare the rainfall data, India Meteorological Department (IMD) calculated the Long Period Average (LPA) of rainfall during period June-August for each of the 16 states. The figure given below shows the actual rainfall (measured in mm) during June-August, 2019 and the percentage deviations from LPA of respective states in 2018. Each state along with its actual rainfall is presented in the figure.

Actual rainfall from June to August 2019

Question (1)

If a 'Heavy Monsoon State' is defined as a state with actual rainfall from June-August, 2019 of 900 mm or more, then approximately what percentage of 'Heavy Monsoon States' have a negative deviation from respective LPAs in 2019?

Handwritten arrow points to the 900 mm mark on the X-axis.

VIDEO SOLUTION

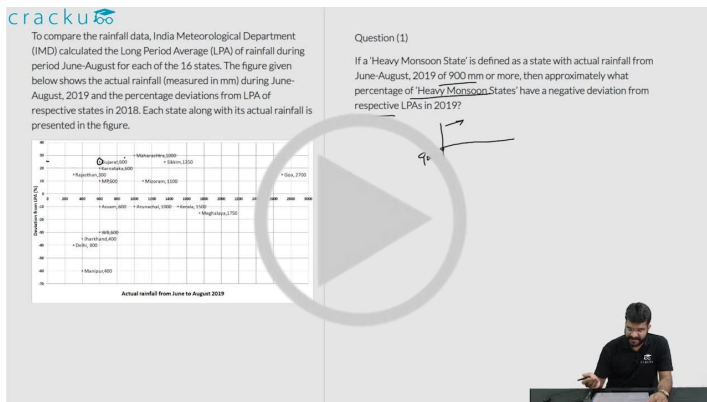
Question 48

What is the average rainfall of all states that have actual rainfall of 600 mm or less in 2019 and have a negative deviation from LPA?

- A 367 mm
- B 500 mm

C 450 mm

D 460 mm



VIDEO SOLUTION

100+ CAT DILR Questions



Question 49

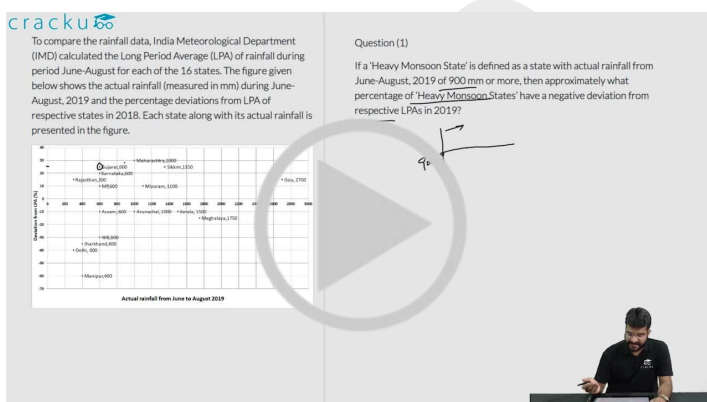
If a 'Heavy Monsoon State' is defined as a state with actual rainfall from June-August, 2019 of 900 mm or more, then approximately what percentage of 'Heavy Monsoon States' have a negative deviation from respective LPAs in 2019?

A 42.86

B 75.00

C 57.14

D 14.29



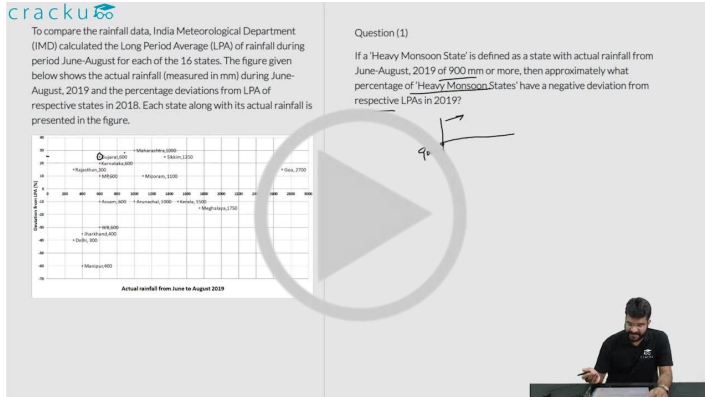
VIDEO SOLUTION

Question 50

The LPA of a state for a year is defined as the average rainfall in the preceding 10 years considering the period of June-August. For example, LPA in 2018 is the average rainfall during 2009-2018 and LPA in 2019 is the average rainfall during 2010-2019. It is also observed that the actual rainfall in Gujarat in 2019 is 20% more than

the rainfall in 2009. The LPA of Gujarat in 2019 is closest to

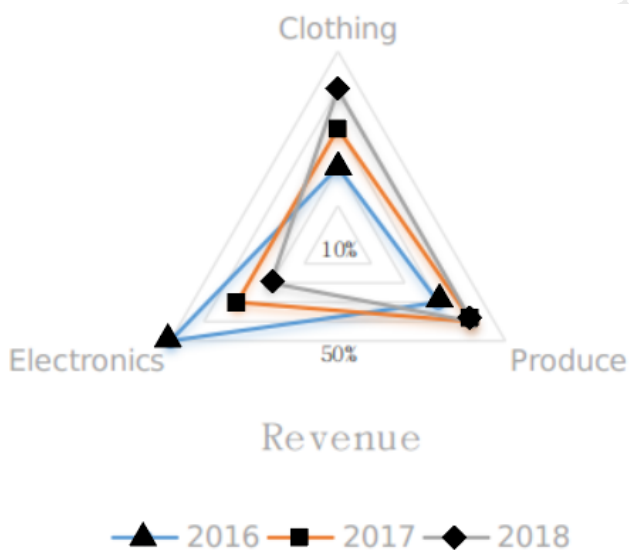
- A 475 mm
- B 505 mm
- C 490 mm
- D 525 mm



VIDEO SOLUTION

Instructions

A large store has only three departments, Clothing, Produce, and Electronics. The following figure shows the percentages of revenue and cost from the three departments for the years 2016, 2017 and 2018. The dotted lines depict percentage levels. So for example, in 2016, 50% of store's revenue came from its Electronics department while 40% of its costs were incurred in the Produce department





In this setup, Profit is computed as (Revenue - Cost) and Percentage Profit as $\text{Profit}/\text{Cost} \times 100\%$.

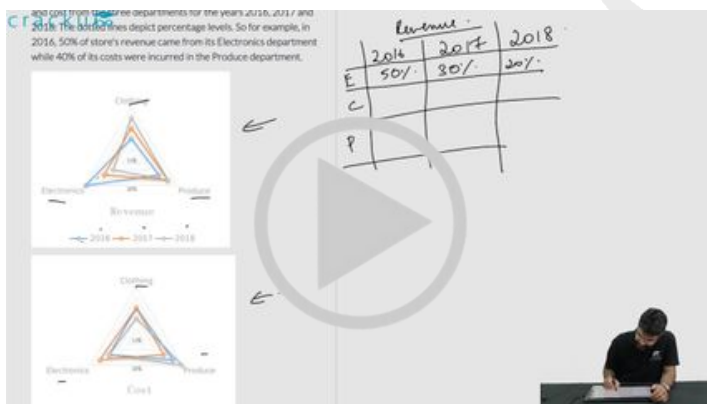
It is known that

1. The percentage profit for the store in 2016 was 100%.
2. The store's revenue doubled from 2016 to 2017, and its cost doubled from 2016 to 2018.
3. There was no profit from the Electronics department in 2017.
4. In 2018, the revenue from the Clothing department was the same as the cost incurred in the Produce department.

Question 51

What was the ratio of revenue generated from the Produce department in 2017 to that in 2018?

- A 16 : 9
- B 4 : 3
- C 9 : 16
- D 8 : 5



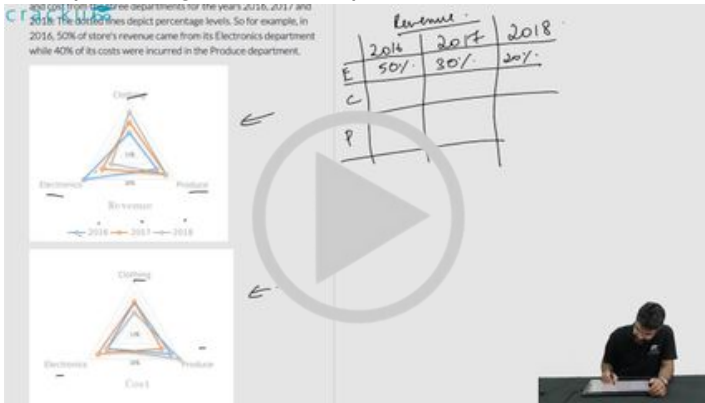
VIDEO SOLUTION

100+ CAT VARC Questions



Question 52

What percentage of the total profits for the store in 2016 was from the Electronics department?

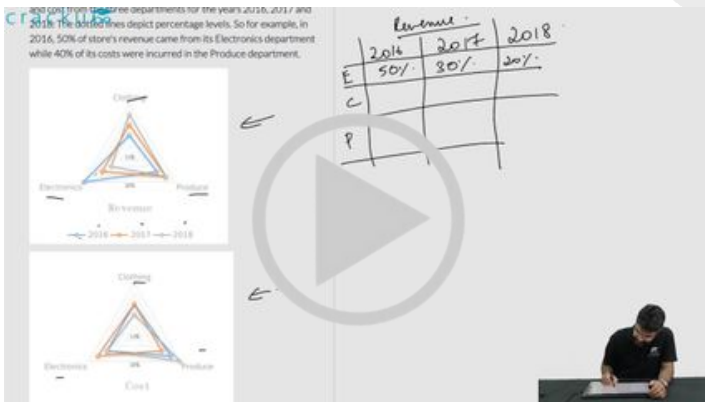


VIDEO SOLUTION

Question 53

What was the approximate difference in profit percentages of the store in 2017 and 2018?

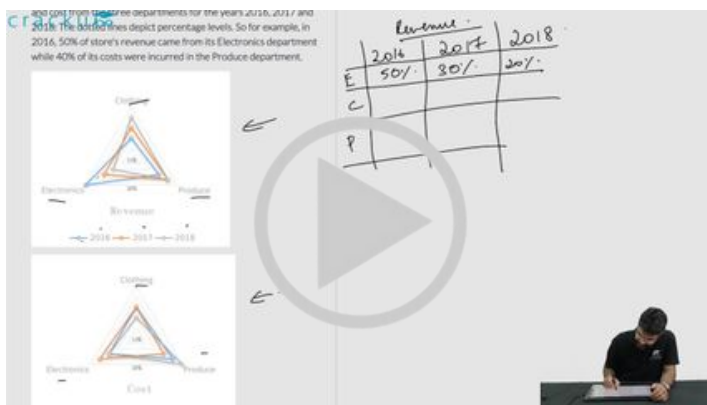
- A 15.5
- B 25.0
- C 8.3
- D 33.3



VIDEO SOLUTION

Question 54

What was the percentage profit of the store in 2018?



VIDEO SOLUTION

CAT Expected Questions PDF



Answers

1.C	2.4	3.C	4.A	5.D	6.E	7.C	8.B
9.D	10.A	11.C	12.C	13.C	14.D	15.A	16.A
17.B	18.D	19.D	20.C	21.D	22.B	23.B	24.B
25.A	26.C	27.C	28.D	29.B	30.C	31.B	32.D
33.B	34.D	35.D	36.A	37.A	38.C	39.D	40.C
41.D	42.B	43.6	44.B	45.D	46.2	47.A	48.D
49.A	50.C	51.D	52.70	53.C	54.25		

Explanations

1.C

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

Daily loss for a share is defined as $(\text{Opening price} - \text{Closing price}) / (\text{Opening price})$

Calculating this for the options:

Stock A: $400/2200=2/11$

Stock B: $300/2000=3/20$

Stock F: $200/1800=1/9$

Stock G gained money that day

Hence Stock A has the highest Daily Loss.

 VIDEO SOLUTION

2.4

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

We are given that, Daily Share Price Variability (SPV) is defined as (Day's high price - Day's low price) / (Average of the opening and closing prices during the day)

Calculating it for the stocks

Stock A: $1200/2000=3/5$

Stock B: $600/1850=60/185$

Stock C: $600/1000=3/5$

Stock D: $900/750=90/75=6/5$

Stock E: $300/1200=1/4$

Stock F: $800/1700=8/17$

Stock G: $900/1450=90/145$

We need to check for stocks greater than 0.5 on that day,

Stock A, Stock C, Stock D, Stock G have SPV greater than 0.5 that day.

Hence, the answer is 4.

 VIDEO SOLUTION

3.C

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

We are given that, Daily Share Price Variability (SPV) is defined as (Day's high price - Day's low price) / (Average of the opening and closing prices during the day)

Calculating it for the four options,

Stock F: $800/1700=8/17$

Stock A: $1200/2000=3/5$

Stock D: $900/750=90/75=6/5$

Stock C: $600/1000=3/5$

Clearly Stock D has the highest SPV.

[VIDEO SOLUTION](#)

4. A

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

There are three bullish shares, C D and G

Lets say a trader buys one share of each of these stocks, and sells them at their day's high

One share of C at opening is 800, sells at 1400

One share of D at opening is 500, 1200

One share of G at opening is 1200, 1900

Total Investment is 2500, and total money after selling is 4500

That is an 80% return since, $\frac{(4500 - 2500)}{2500} = 0.8$

[VIDEO SOLUTION](#)

5. D

We know the following relation that , revenue = volume * ARDT . To have equal value the ratio of revenue by ARDT should be constant.

Considering UK we have volume = 12.5/30 and of Spain we have 6.5/15 which is fairly constant as compared to others.

Hence option D.

[VIEW SOLUTION](#)

 **CAT Important Topics** 

6. E

If the volume of data transfer in India is the same as that of Singapore then we have $(R_i * 7.5 / 0.9) = (20 * R_s / 9.5)$; we get R_s nearly equal to $4 * R_i$.

Hence Total revenue in Singapore is about 4 time that of India.

[VIEW SOLUTION](#)

7. C

Revenue from data transfer as a percentage of total revenue will triple for India and double for Sweden.

Also the total revenue in India is twice that of Sweden.

Taking volume constant we have for India $3 * 0.9 * R_i = ARDT * vol$

For Sweden $0.18 * 2 * R_s = 6 * Vol$

Since vol is constant and $R_i = 2 * R_s$ we have $ARDT = 9$.

Thus % increase in $ARDT = 800\%$.

[VIEW SOLUTION](#)

8. B

The data can be tabulated as follows(approximately):

Parameters	Vendor 1	Vendor 2	Vendor 3	Vendor 4	Vendor 5
Reliability	52	40	75	26	60
Reach	80	58	63	46	70
Quality	72	69	62	40	48
Features	40	45	56	90	75
Customer Services	55	41	50	70	28
Cost	77	81	90	71	50

Top 3 on Reliability: Vendor 3, Vendor 5

Top 3 on Reach: Vendor 1, Vendor 5

Top 3 on Quality: Vendor 1, Vendor 2

Top 3 on Features: Vendor 4, Vendor 5

Top 3 on Customer Services: Vendor 4, Vendor 1

Top 3 on Cost: Vendor 3, Vendor 2

Vendor 1: 3 times Vendor 2: Only once Vendor 3: 2 times Vendor 4: 2 times Vendor 5: 3 times

Here 1 and 5 comes 3 times. Hence B is the answer.

[VIDEO SOLUTION](#)

9. D

The data can be tabulated as follows(approximately):

Parameters	Vendor 1	Vendor 2	Vendor 3	Vendor 4	Vendor 5
Reliability	52	40	75	26	60
Reach	80	58	63	46	70
Quality	72	69	62	40	48
Features	40	45	56	90	75
Customer Services	55	41	50	70	28
Cost	77	81	90	71	50
Total	376	334	396	343	321

The average of the vendor will be highest which has highest total score. Hence vendor 3 has the highest average.

[VIDEO SOLUTION](#)

10. A

The data can be tabulated as follows(approximately):

Parameters	Vendor 1	Vendor 2	Vendor 3	Vendor 4	Vendor 5
Reliability	52	40	75	26	60
Reach	80	58	63	46	70
Quality	72	69	62	40	48
Features	40	45	56	90	75
Customer Services	55	41	50	70	28
Cost	77	81	90	71	50

Customer Services: 28,41,50,55,70 (The median is 50)

Cost: 50,71,77,81,90 (The median is 77)

Reliability: 26, 40, 52, 60, 75 (The median is 52)

Quality: 40, 48, 62, 69, 72 (The median is 62)

Features: 40, 45, 56, 75, 90 (The median is 56)

Reach: 46, 58, 63, 70, 80 (The median is 63)

Hence the customer services has the lowest median.

[VIDEO SOLUTION](#)



CAT WhatsApp Group



11. C

The data can be tabulated as follows (approximately):

Parameters	Vendor 1	Vendor 2	Vendor 3	Vendor 4	Vendor 5
Reliability	52	40	75	26	60
Reach	80	58	63	46	70
Quality	72	69	62	40	48
Features	40	45	56	90	75
Customer Services	55	41	50	70	28
Cost	77	81	90	71	50

Top 3 on Reliability: Vendor 3, Vendor 5, Vendor 1

Top 3 on Reach: Vendor 1, Vendor 5, Vendor 3

Top 3 on Quality: Vendor 1, Vendor 2, Vendor 3

Top 3 on Features: Vendor 4, Vendor 5, Vendor 3

Top 3 on Customer Services: Vendor 4, Vendor 1, Vendor 3

Top 3 on Cost: Vendor 3, Vendor 2, Vendor 1

Only Vendor 3 ranks among top 3 in all the six parameters.

[VIDEO SOLUTION](#)

12. C

Total sales = $150 \times 40 = 6000$. Total costs = $2000 + 3650$ (approx). = 5600 to 5700. Hence, profit = $6000 - 5600 = 400$ (approx). Hence option C.

[VIEW SOLUTION](#)

13. C

For 40 units total fixed costs would be $1200 + 800 = 2000$ and variable cost = 3700. So total cost = 5700 (approx). Hence approximate average unit cost for July = $5700 / 40 = 142.5$. Hence option C is the correct answer.

[VIEW SOLUTION](#)

14. D

When we take some value and check for the variation, we find that actual variation is opposite to that of option C.

Upto a certain level, AC is greater than MC and then AC is less than MC.

For producing 31 units, the average cost is $1200 + 800 + 2400 = \text{Rs. } 4400$.

Average cost = $4400/31 = \text{Rs. } 142$ (Approx).

Let us calculate the marginal cost at this point.

As the number of units increases from 31 to 40, the variable cost increases from Rs.2400 to Rs.3700.

Marginal cost = $(3700-2400)/10 = 1300/10 = \text{Rs. } 130$.

$MC < AC$ for 31 units.

Let us calculate MC and AC for 60 units.

Average cost = $(2000 + 6800)/60 = 8800/60 = \text{Rs. } 146.66$

As the number of units increases from 50 to 60, the variable cost increases from Rs.5000 to Rs.6800.

Marginal cost = $\text{Rs. } 1800/10 = \text{Rs. } 180$

$MC > AC$ for 60 units.

As we can see, the trend is the inverse of what is mentioned in option C. Since none of the options capture the trend precisely, option D is the right answer.

[VIEW SOLUTION](#)

15. A

As adding a shift dramatically increases the costs, we need to consider the two regions, $0 < x \leq 30$ and $30 < x \leq 60$, separately.

Region 1: As variable cost is less than equal to 100, and sales price is 150, profit would be max when production is maximum possible i.e. at 30 units.

Hence, at 30 units, total cost = $800 + 2400 = 3200$. Sales = $30 * 150 = 4500$. Profit = $4500 - 3200 = 1300$

Region 2: As variable cost is less than equal to 100, and sales price is 150, profit would be max when production is maximum possible i.e. at 60 units

Hence, at 60 units, total cost = $800 + 1200 + 6500 = 8500$. Sales = $150 * 60 = 9000$. Hence, Profit = $9000 - 8500 = 500$

Hence, profit is maximum at 30 units.

[VIEW SOLUTION](#)

A promotional banner for CAT Mock Tests. On the left, there is a photo of two people, a man and a woman, both wearing black t-shirts with 'cracku' written on them. To their right, the text 'CAT Mock Test' is written in large, bold, blue letters. Below this, a yellow box contains the text 'Mocks designed exactly like the actual CAT'. On the far right, there is a small logo for 'cracku' and an illustration of a desk with a lamp, a clock, and some books, with the word 'MOCK' written above it.

16. A

As the graph is an increasing function, MC will always increase for increase in number of units.

[VIEW SOLUTION](#)

17. B

We know that MC is the rate of change in total cost for unit change in quantity produced. SO for 40 units total cost = $2000 + 3700 = 5700$ and for 50 units = $2000 + 5000 = 7000$. So for additional 10 units change in total costs is 1300 . So for 1 unit we have $MC = 130$.

[VIEW SOLUTION](#)

18. D

The fertilizer and food production has been inversely proportional over the years. ie when food production increased, fertilizer production fell and vice versa. But as we can see in 1988-89, when food production remained constant, the fertilizer production fell steeply and in 89-90 when food production increased steeply, the fertilizer production stayed constant. So, 1989 is the year of the anomaly.

[VIEW SOLUTION](#)

19. D

Just by looking at the graphs of the previous year's it is difficult to guess how the food production in the year 1992 will be.

This is especially true as we don't have any information about the fertilizer production in the year 1992.

Hence, the correct answer is option (d)

[VIEW SOLUTION](#)

20. C

From the graph, in the year 1988, the ratio of food production to fertilizer production is 5:3.5 i.e 10:7

Let the food production be $10X$ and the fertilizer production be $7X$

Hence, the total is $10X + 7X = 17X$ which equals 170

So, $X = 10$ and the total food production equals $10 \times 10 = 100$

[VIEW SOLUTION](#)



21. D

In order to find the number of years for which the sum is same, we need to determine the change in production of food and fertilizer. If there is no effective change, then the sum of their productions is same.

Change in 1984 $\Rightarrow +1 - 1.5 = -0.5$

Change in 1985 $\Rightarrow 0$

Change in 1986 $\Rightarrow -1.5 + 1.5 = 0$

Change in 1987 $\Rightarrow 0$

Change in 1988 $\Rightarrow +1.5 - 1.5 = 0$

Change in 1989 $\Rightarrow -2.5 + 0 = -2.5$

Change in 1990 $\Rightarrow 0 + 1.5 = 1.5$

Change in 1991 $\Rightarrow 0$

The change is 0 in 5 years. Hence, the answer is option D.

[VIEW SOLUTION](#)

22. B

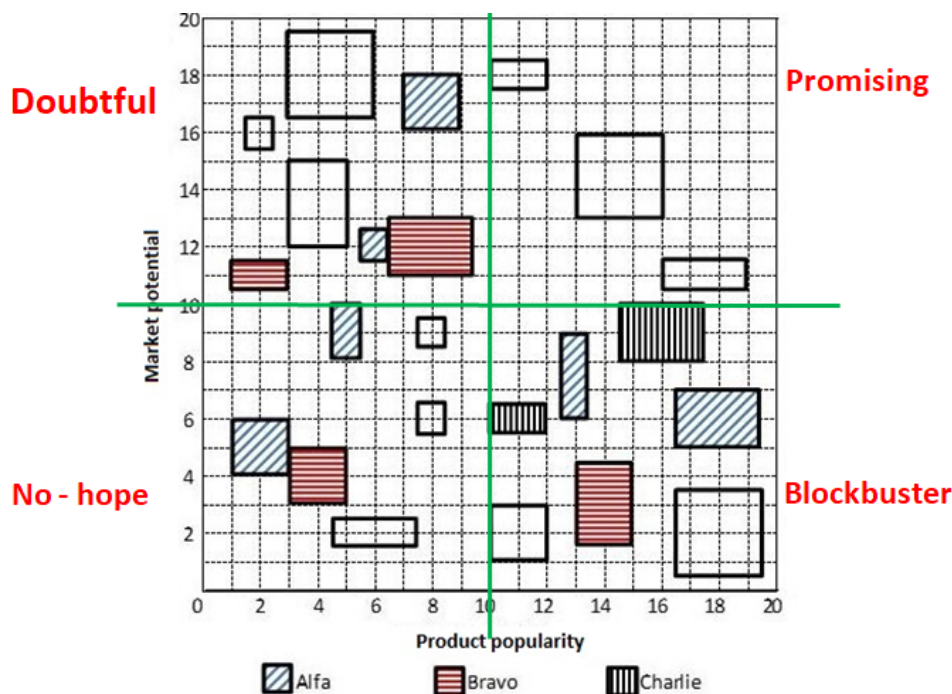
Let each unit in the scale be equal to X

So, fertilizer production in 1983 is 2.5X
 So, fertilizer production in 1984 is 3.5X
 So, fertilizer production in 1985 is 3.5X
 So, fertilizer production in 1986 is 2X
 So, fertilizer production in 1987 is 2X
 So, fertilizer production in 1988 is 3.5X
 Assuming fertilizer production in 1989 is same as that of 1988 it equals 3.5X
 So, fertilizer production in 1990 is X
 So, fertilizer production in 1991 is X
 Hence, the total is 22.5X
 But this equals 450 million tonnes. Hence, $X = 20$ million tonnes
 Therefore, food production in 1983 = 6.5X equals 130 million tonnes

[VIEW SOLUTION](#)

23. B

Let us divide the given figure in four quadrants (Q1, Q2, Q3, Q4).



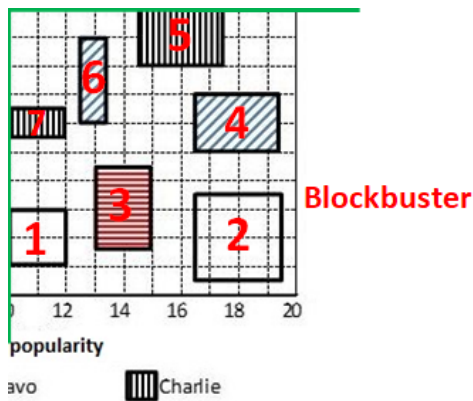
Let us solve this problem by considering only one category at a time.

(A) Blockbuster category: We have two information regarding Blockbuster category.

1. Alfa and Bravo had the same number of products in the Blockbuster category.

There are a total of 7 products in Blockbuster category. Alfa already has two products in blockbuster category. If Alfa has 3 products in blockbuster category then Bravo will also have 3 products in Blockbuster category which is not possible as there are 2 products of Charlie. Hence, we can say that Alfa and Bravo have 2 products each in Blockbuster category whereas Charlie has 3 products in Blockbuster category.

2. It is also given that Charlie had a higher revenue than Bravo from products in the Blockbuster category.



We know that one of the product 1 and 2 is from Charlie and the other is from Bravo.

If product 1 is from Charlie, then we can say that products 1, 7 and 5 are from Charlie. Therefore, revenue generated by products in Charlie category = $2 + 4 + 6 = 12$ units. (Assuming area of a square to be 1 unit)

In this case product 2 and product 4 are from Bravo. Therefore, revenue generated by products in Bravo category = $6 + 9 = 15$ units.

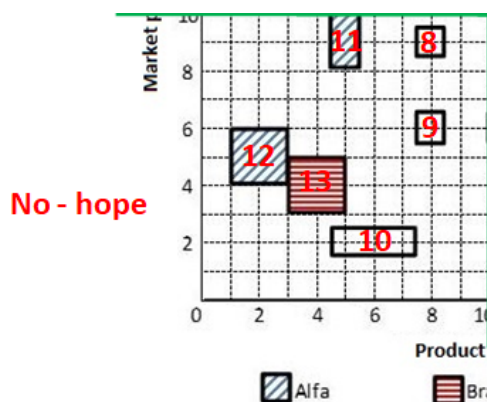
We can see that products from Charlie have a higher revenue than Bravo. Hence, this case is not possible.

Therefore, we can say that Product 1 is from Bravo and Product 2 is from Charlie. We have determined each product's company name in Blockbuster category.

(B) No-hope category: We have two information regarding No-hope category.

(1) Charlie had more products than Bravo but fewer products than Alfa in the No-hope category. Bravo and Charlie had the same revenue from products in the No-hope category. There are a total of 6 products in no-hope category.

Therefore, we can say that Alfa, Charlie and Bravo have 3, 2 and 1 products in No-hope category in that order.



Bravo and Charlie had the same revenue from products in the No-hope category.

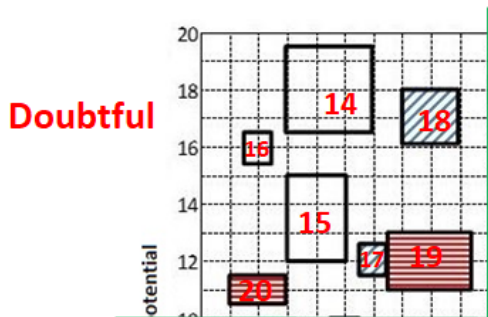
Revenue generated for Bravo in the No-hope category = 4 units. We need same revenue for Charlie which has 2 products in this category. Hence, we can say that Product 10 and one of product 8 and 9 is from Charlie and other is from Alfa. Let's assume that product 8 is from Charlie and product 9 is from Alfa.

(C) Doubtful category: We have two information regarding Doubtful category.

(1). Charlie did not have any product in the Doubtful category, while Alfa had one product more than Bravo in this category.

(2). Bravo had a higher revenue than Alfa from products in the Doubtful category.

We can see that there are a total of 7 products in this category. Hence, we can say that 4 products are from Alfa and 3 products are from Bravo.



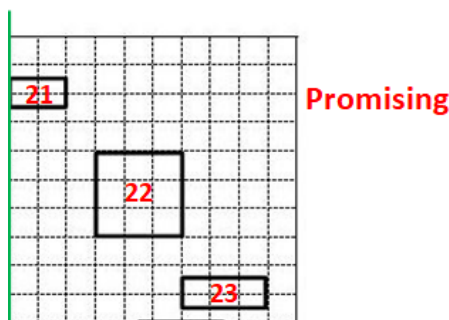
We can see that one of product 14, 15 and 16 is from Bravo company and others are from Alfa company. Bravo will have higher revenue than Alfa only when product no. 14 is from Bravo and others (15 and 16) are from Alfa category.

In this case total revenue by Bravo = Product 14 + Product 19 + Product 20 = 9 + 6 + 2 = 17

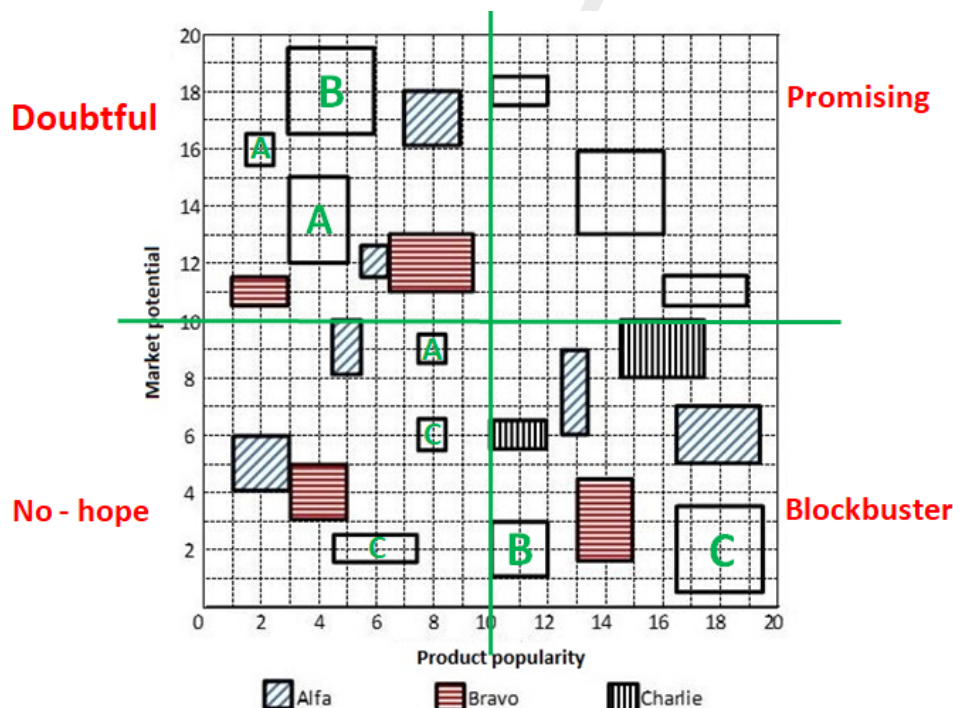
Similarly, total revenue by Charlie = Product 15 + Product 16 + Product 17 + Product 18 = 6 + 1 + 1 + 4 = 12

(D) Promising category: We have only 1 direct information regarding Promising category.

1. Each company had an equal number of products in the Promising category.



There are a total of 3 products in promising category with different revenue. Therefore, we can say that each company had 1 product in promising category. We are given that Alfa and Charlie had the same total revenue considering all products. We can calculate the revenue generated by Alfa and Charlie from the products in categories.



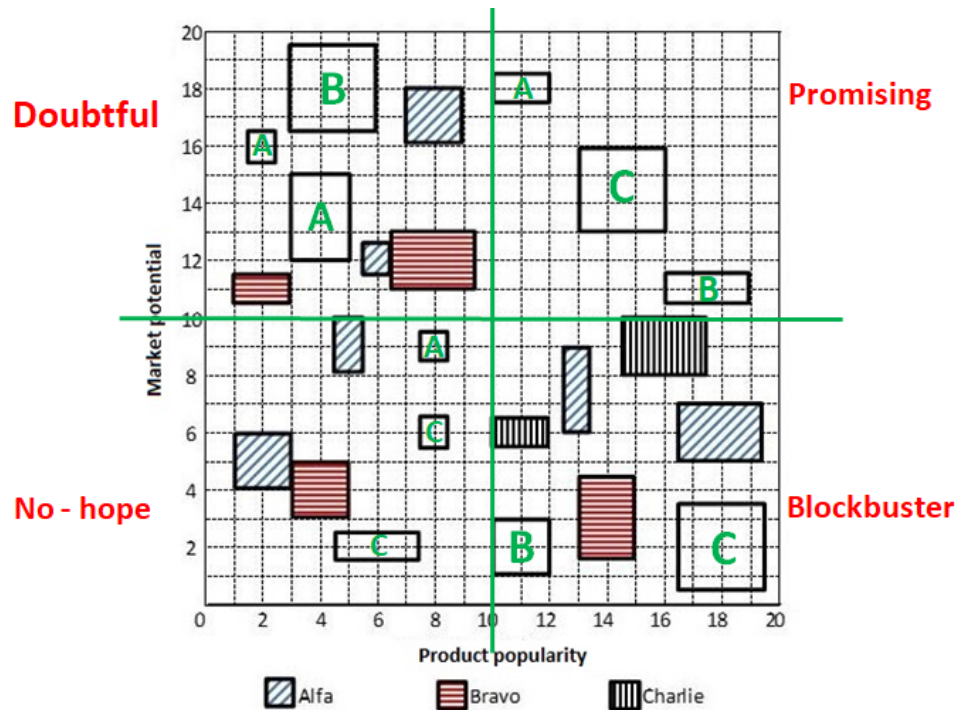
Revenue generated by Charlie from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (9+6+2) + (3+1) + (0) = 21 \text{ units}$$

Revenue generated by Alfa from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (6+3) + (4+2+1) + (1+6+4+1) = 28 \text{ units}$$

We can see the difference between revenue generated by Charlie and Alfa from remaining categories is 7 units. Hence, we can say that Charlie's product's revenue should be 7 units more than Alfa's product's revenue in Promising category. That is possible only in one case where product 22 is from Charlie and product 21 is from Alfa. Consequently, we can say that product 23 is from Bravo. Now we have identified each product's company name we can answer all the questions.



Let us check all options.

Option (A): Alfa's revenue from Blockbuster products was the same as Charlie's revenue from Promising products

Alfa's revenue from Blockbuster products = $6 + 3 = 9$ units.

Charlie's revenue from Promising products = 9 units. Hence, this statement is **true**.

Option (B): Bravo's revenue from Blockbuster products was greater than Alfa's revenue from Doubtful products.

Bravo's revenue from Blockbuster products = $6 + 4 = 10$ units.

Alfa's revenue from Doubtful products = $6 + 4 + 1 + 1 = 12$ units. Hence, this statement is **false**.

Option (C): Bravo and Charlie had the same revenues from No-hope products.

Bravo's revenue from No-hope products = 4 units.

Charlie's revenue from No-hope products = $3 + 1 = 4$ units. Hence, this statement is **true**.

Option (D): The total revenue from No-hope products was less than the total revenue from Doubtful products

Revenue generated by the products in Doubtful category = $1 + 9 + 4 + 6 + 2 + 1 + 6 = 29$ units.

Revenue generated by the products in No-hope category = $4 + 4 + 3 + 2 + 1 + 1 = 15$ units. Hence, this statement is **true**.

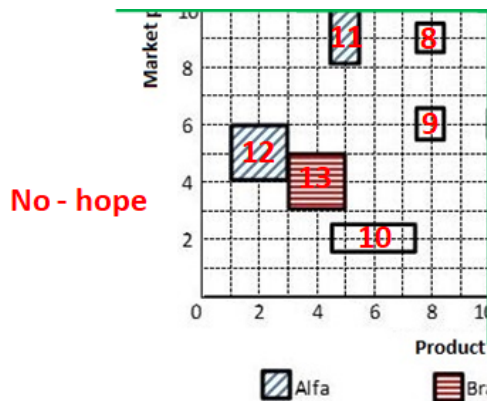
We can see that statement mentioned in option B is false. Therefore, option B is the correct answer.

VIDEO SOLUTION

24. B

Let us divide the given figure in four quadrants (Q1, Q2, Q3, Q4).

Therefore, we can say that Alfa, Charlie and Bravo have 3, 2 and 1 products in No-hope category in that order.



Bravo and Charlie had the same revenue from products in the No-hope category.

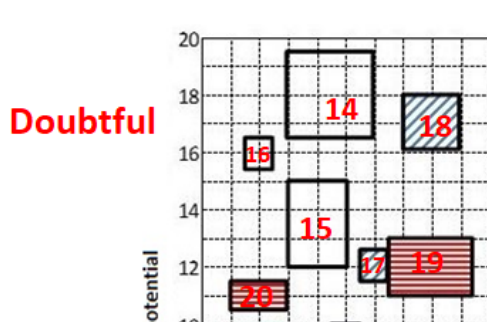
Revenue generated for Bravo in the No-hope category = 4 units. We need same revenue for Charlie which has 2 products in this category. Hence, we can say that Product 10 and one of product 8 and 9 is from Charlie and other is from Alfa. Let's assume that product 8 is from Charlie and product 9 is from Alfa.

(C) Doubtful category: We have two information regarding Doubtful category.

(1). Charlie did not have any product in the Doubtful category, while Alfa had one product more than Bravo in this category .

(2). Bravo had a higher revenue than Alfa from products in the Doubtful category.

We can see that there are a total of 7 products in this category. Hence, we can say that 4 products are from Alfa and 3 products are from Bravo.



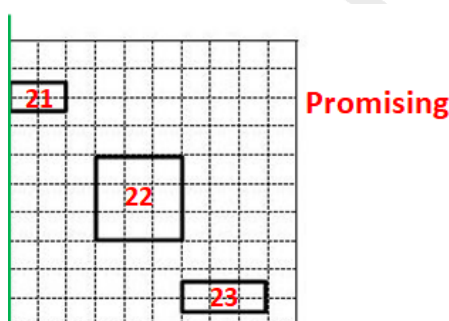
We can see that one of product 14, 15 and 16 is from Bravo company and others are from Alfa company. Bravo will have higher revenue than Alfa only when product no. 14 is from Bravo and others (15 and 16) are from Alfa category.

In this case total revenue by Bravo = Product 14 + Product 19 + Product 20 = 9 + 6 + 2 = 17

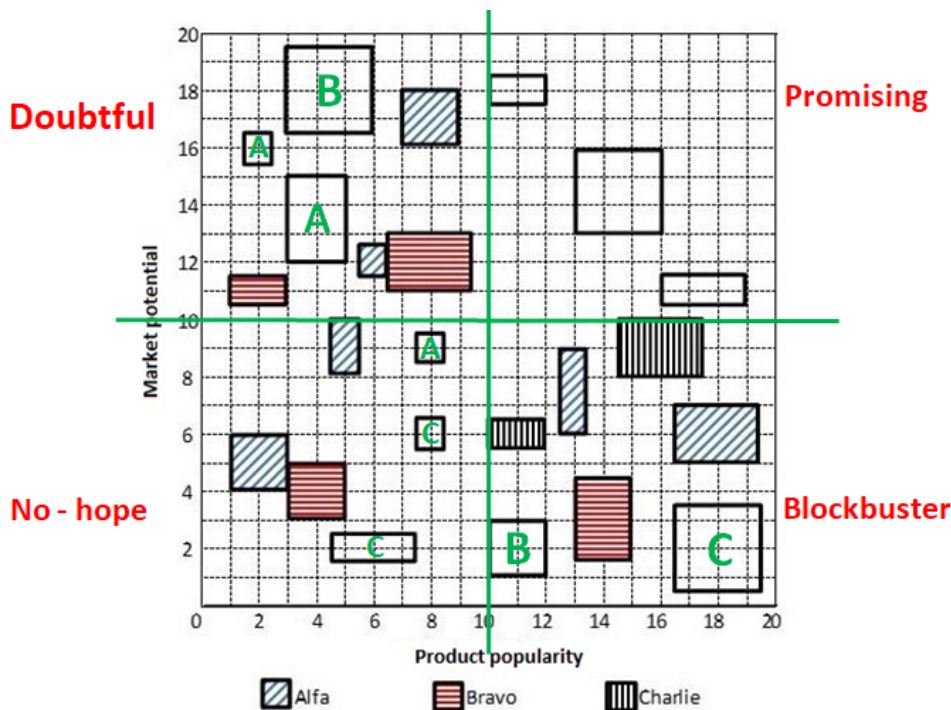
Similarly, total revenue by Charlie = Product 15 + Product 16 + Product 17 + Product 18 = 6 + 1 + 1 + 4 = 12

(D) Promising category: We have only 1 direct information regarding Promising category.

1. Each company had an equal number of products in the Promising category.



There are a total of 3 products in promising category with different revenue. Therefore, we can say that each company had 1 product in promising category. We are given that Alfa and Charlie had the same total revenue considering all products. We can calculate the revenue generated by Alfa and Charlie from the products in categories.



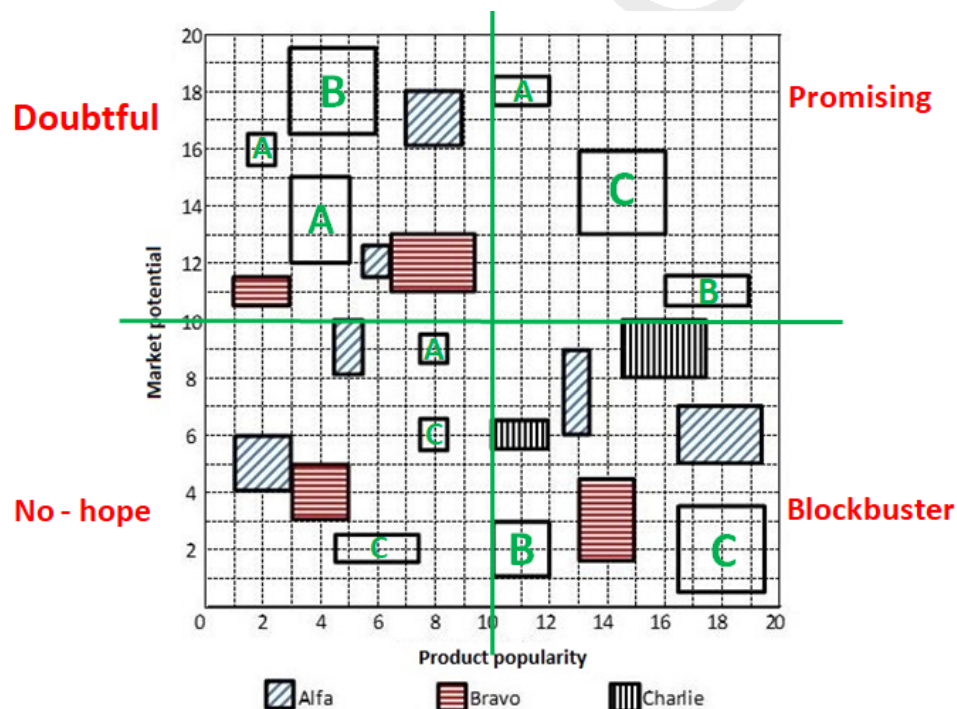
Revenue generated by Charlie from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (9+6+2) + (3+1) + (0) = 21 \text{ units}$$

Revenue generated by Alfa from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (6+3) + (4+2+1) + (1+6+4+1) = 28 \text{ units}$$

We can see the difference between revenue generated by Charlie and Alfa from remaining categories is 7 units. Hence, we can say that Charlie's product's revenue should be 7 units more than Alfa's product's revenue in Promising category. That is possible only in one case where product 22 is from Charlie and product 21 is from Alfa. Consequently, we can say that product 23 is from Bravo. Now we have identified each product's company name we can answer all the questions.



Revenue generated by the products in Promising category = 2 + 9 + 3 = 14 units.

Revenue generated by the products in Doubtful category = 1 + 9 + 4 + 6 + 2 + 1 + 6 = 29 units.

Revenue generated by the products in No-hope category = $4 + 4 + 3 + 2 + 1 + 1 = 15$ units.

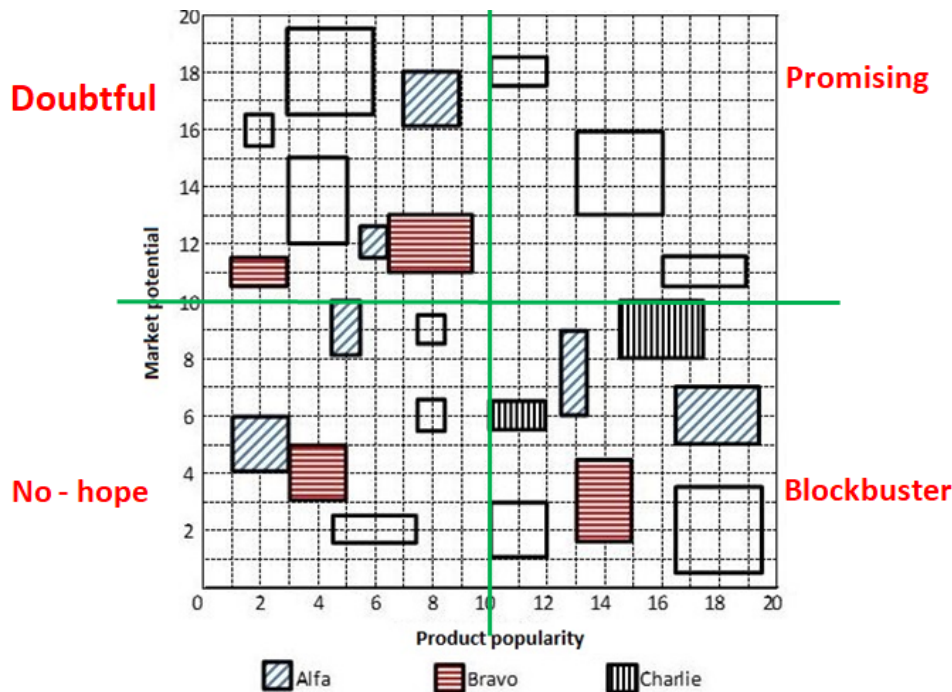
Revenue generated by the products in Blockbuster category = $6 + 3 + 6 + 2 + 4 + 6 + 9 = 36$ units.

We can see that the revenue generated is the highest for Blockbuster category. Hence, option B is the correct answer.

VIDEO SOLUTION

25. A

Let us divide the given figure in four quadrants (Q1, Q2, Q3, Q4).



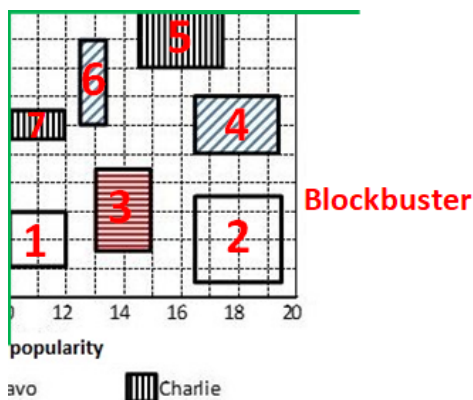
Let us solve this problem by considering only one category at a time.

(A) Blockbuster category: We have two information regarding Blockbuster category.

1. Alfa and Bravo had the same number of products in the Blockbuster category.

There are a total of 7 products in Blockbuster category. Alfa already has two products in blockbuster category. If Alfa has 3 products in blockbuster category then Bravo will also have 3 products in Blockbuster category which is not possible as there are 2 products of Charlie. Hence, we can say that Alfa and Bravo have 2 products each in Blockbuster category whereas Charlie has 3 products in Blockbuster category.

2. It is also given that Charlie had a higher revenue than Bravo from products in the Blockbuster category.



We know that one of the product 1 and 2 is from Charlie and the other is from Bravo.

If product 1 is from Charlie, then we can say that products 1, 7 and 5 are from Charlie. Therefore, revenue generated by products in Charlie category = $2 + 4 + 6 = 12$ units. (Assuming area of a square to be 1 unit)

In this case product 2 and product n are from Bravo. Therefore, revenue generated by products in Bravo category = $6 + 9 = 15$ units.

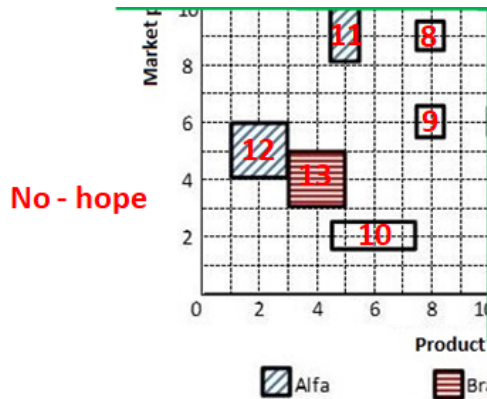
We can see that products from Charlie have a higher revenue than Bravo. Hence, this case is not possible.

Therefore, we can say that Product 1 is from Bravo and Product 2 is from Charlie. We have determined each product's company name in Blockbuster category.

(B) No-hope category: We have two information regarding No-hope category.

(1) Charlie had more products than Bravo but fewer products than Alfa in the No-hope category. Bravo and Charlie had the same revenue from products in the No-hope category. There are a total of 6 products in no-hope category.

Therefore, we can say that Alfa, Charlie and Bravo have 3, 2 and 1 products in No-hope category in that order.



Bravo and Charlie had the same revenue from products in the No-hope category.

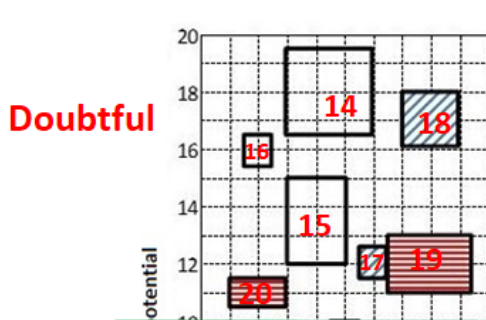
Revenue generated for Bravo in the No-hope category = 4 units. We need same revenue for Charlie which has 2 products in this category. Hence, we can say that Product 10 and one of product 8 and 9 is from Charlie and other is from Alfa. Let's assume that product 8 is from Charlie and product 9 is from Alfa.

(C) Doubtful category: We have two information regarding Doubtful category.

(1). Charlie did not have any product in the Doubtful category, while Alfa had one product more than Bravo in this category.

(2). Bravo had a higher revenue than Alfa from products in the Doubtful category.

We can see that there are a total of 7 products in this category. Hence, we can say that 4 products are from Alfa and 3 products are from Bravo.



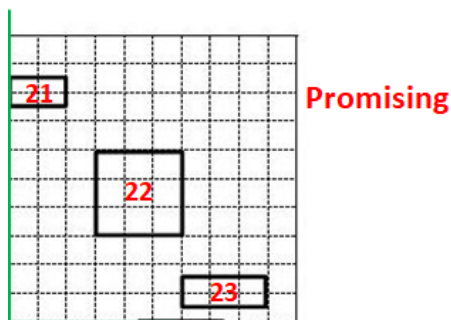
We can see that one of product 14, 15 and 16 is from Bravo company and others are from Alfa company. Bravo will have higher revenue than Alfa only when product no. 14 is from Bravo and others (15 and 16) are from Alfa category.

In this case total revenue by Bravo = Product 14 + Product 19 + Product 20 = 9 + 6 + 2 = 17

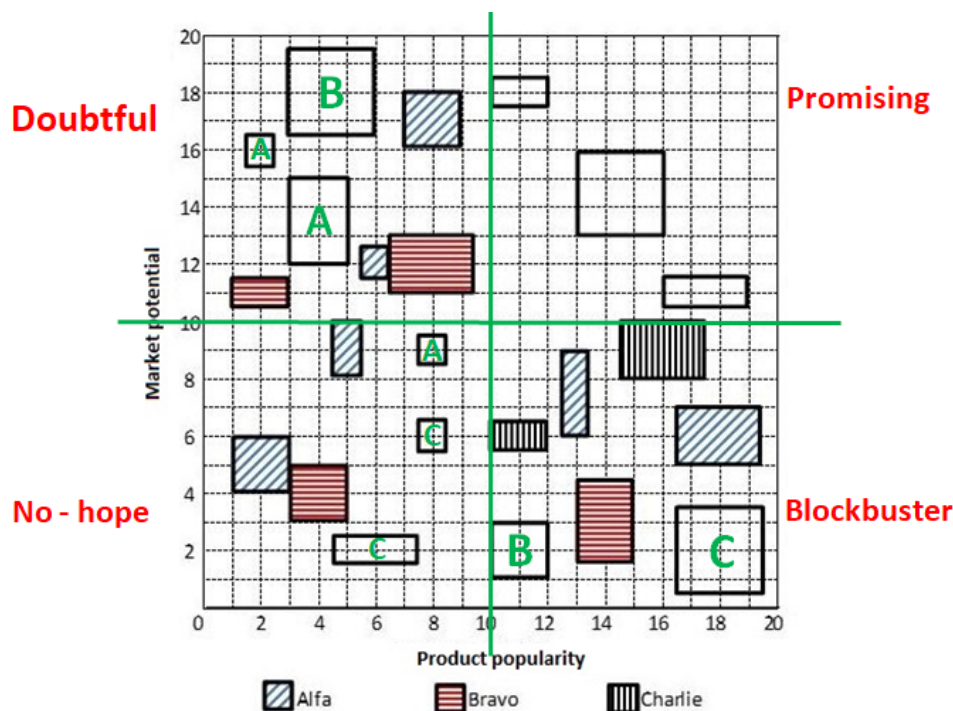
Similarly, total revenue by Charlie = Product 15 + Product 16 + Product 17 + Product 18 = 6 + 1 + 1 + 4 = 12

(D) Promising category: We have only 1 direct information regarding Promising category.

1. Each company had an equal number of products in the Promising category.



There are a total of 3 products in promising category with different revenue. Therefore, we can say that each company had 1 product in promising category. We are given that Alfa and Charlie had the same total revenue considering all products. We can calculate the revenue generated by Alfa and Charlie from the products in categories.



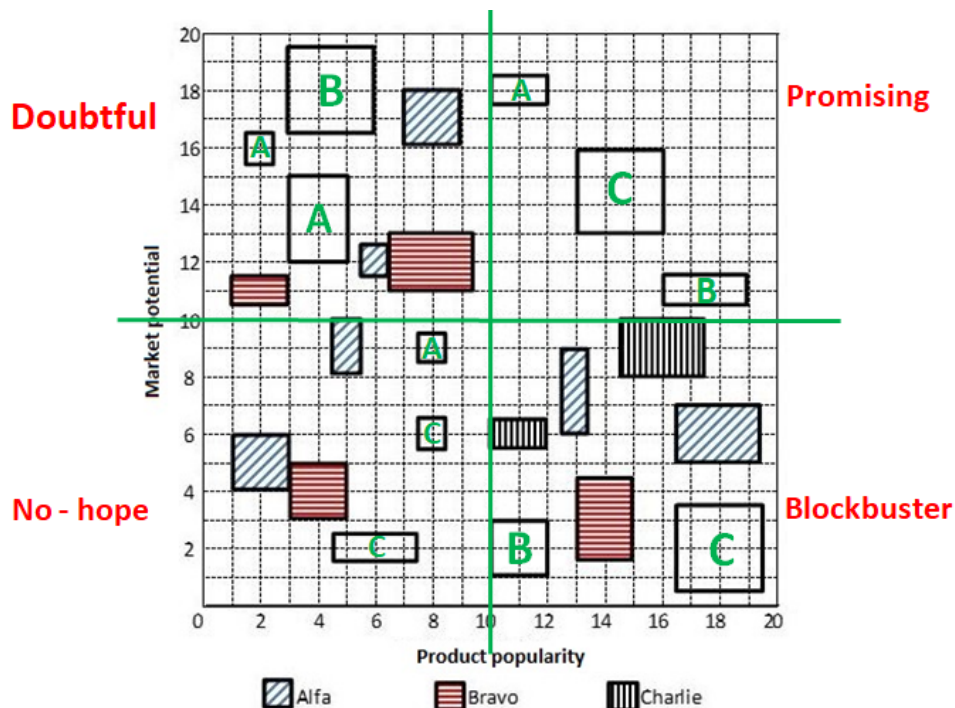
Revenue generated by Charlie from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (9+6+2) + (3+1) + (0) = 21 \text{ units}$$

Revenue generated by Alfa from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (6+3) + (4+2+1) + (1+6+4+1) = 28 \text{ units}$$

We can see the difference between revenue generated by Charlie and Alfa from remaining categories is 7 units. Hence, we can say that Charlie's product's revenue should be 7 units more than Alfa's product's revenue in Promising category. That is possible only in one case where product 22 is from Charlie and product 21 is from Alfa. Consequently, we can say that product 23 is from Bravo. Now we have identified each product's company name we can answer all the questions.



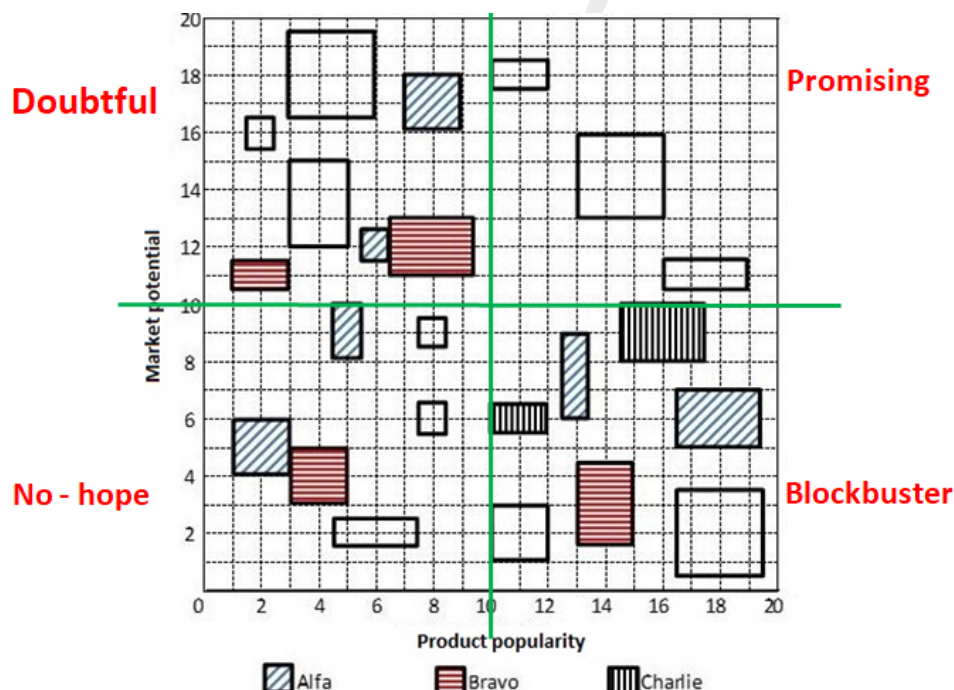
Bravo had in No-hope, Doubtful, Promising and Blockbuster categories respectively = 1, 3, 1, 2. Hence, option A is the correct answer.

VIDEO SOLUTION



26. C

Let us divide the given figure in four quadrants (Q1, Q2, Q3, Q4).

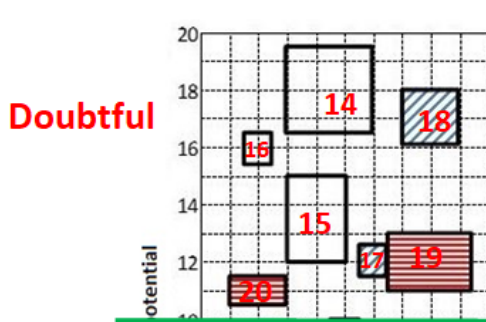


Let us solve this problem by considering only one category at a time.

(A) Blockbuster category: We have two information regarding Blockbuster category.

1. Alfa and Bravo had the same number of products in the Blockbuster category.

We can see that there are a total of 7 products in this category. Hence, we can say that 4 products are from Alfa and 3 products are from Bravo.



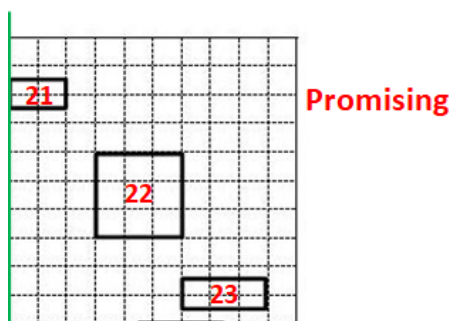
We can see that one of product 14, 15 and 16 is from Bravo company and others are from Alfa company. Bravo will have higher revenue than Alfa only when product no. 14 is from Bravo and others (15 and 16) are from Alfa category.

In this case total revenue by Bravo = Product 14 + Product 19 + Product 20 = $9 + 6 + 2 = 17$

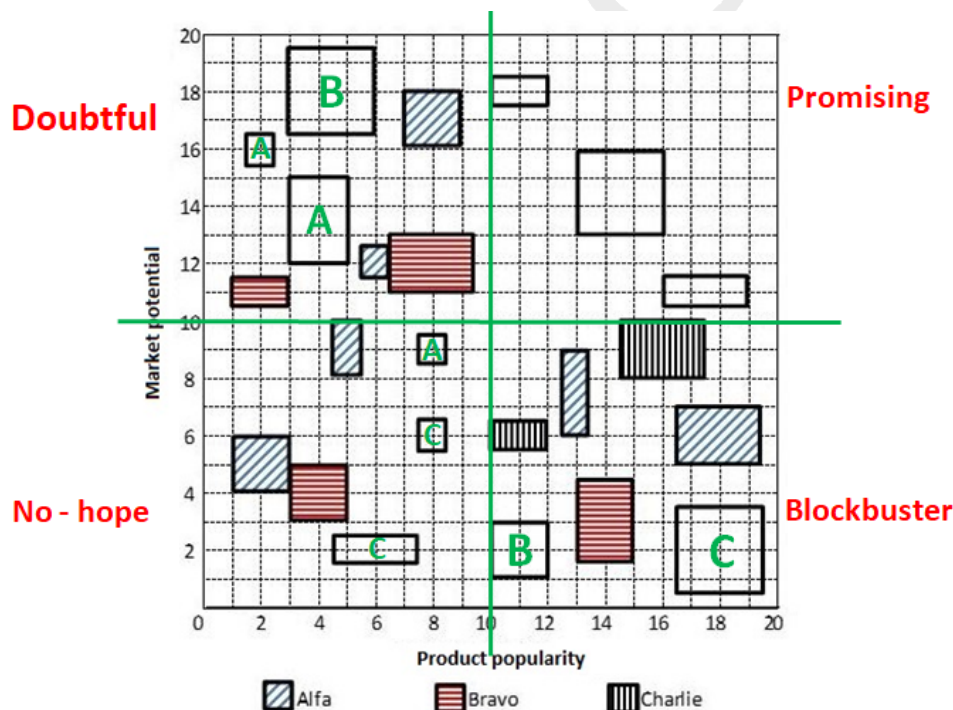
Similarly, total revenue by Charlie = Product 15 + Product 16 + Product 17 + Product 18 = $6 + 1 + 1 + 4 = 12$

(D) Promising category: We have only 1 direct information regarding Promising category.

1. Each company had an equal number of products in the Promising category.



There are a total of 3 products in promising category with different revenue. Therefore, we can say that each company had 1 product in promising category. We are given that Alfa and Charlie had the same total revenue considering all products. We can calculate the revenue generated by Alfa and Charlie from the products in categories.



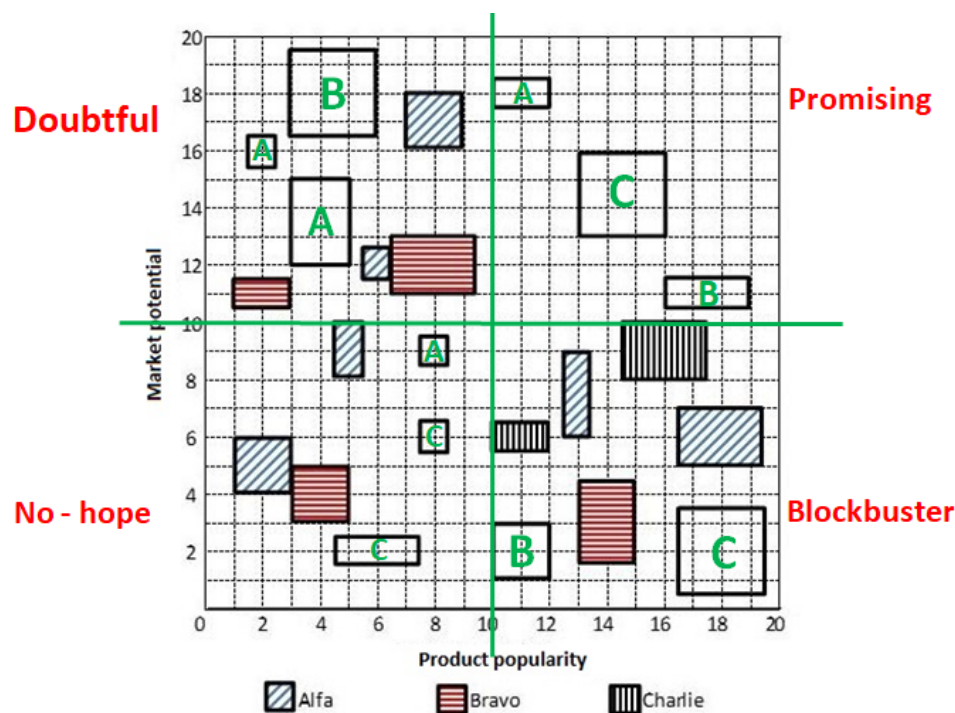
Revenue generated by Charlie from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (9+6+2) + (3+1) + (0) = 21 \text{ units}$$

Revenue generated by Alfa from all categories except Promising = From Blockbuster + From No-hope + From Doubtful

$$\Rightarrow (6+3) + (4+2+1) + (1+6+4+1) = 28 \text{ units}$$

We can see the difference between revenue generated by Charlie and Alfa from remaining categories is 7 units. Hence, we can say that Charlie's product's revenue should be 7 units more than Alfa's product's revenue in Promising category. That is possible only in one case where product 22 is from Charlie and product 21 is from Alfa. Consequently, we can say that product 23 is from Bravo. Now we have identified each product's company name we can answer all the questions.



Total revenue generated by Bravo products alone = From Blockbuster + From No-hope + From Doubtful + From Promising

$$\Rightarrow (6+4) + (4) + (9+6+2) + (3) = 34 \text{ units}$$

One box is equivalent to Rs. 1 crore therefore, we can say that total revenue generated by Bravo = Rs. 34 crores. Hence, option C is the correct answer.

[VIDEO SOLUTION](#)

27. **C**

Take a look at the table,

The low producing areas corresponding to Crop 1 is R9, R10, R11

Now medium yield area corresponding to Crop 3 is R9 and high yield are corresponding to crop 4 is R10 and R11.

Hence all three area are mentioned in the above two categories.

[VIEW SOLUTION](#)

28. **D**

High quality crop 2 is produced by R5, R8 and R12.

None of these regions produce crop 3.

Hence, option D is true.

None of the other options are true.

[VIEW SOLUTION](#)

29. **B**

It can be seen that region 9 produce medium qualities of Crop-1 or Crop-2 and also produce low quality of Crop-3 or Crop-4.

[VIEW SOLUTION](#)

30. **C**

The values given in both charts together are represented in the table below, with the import tariff percentages charged by each of the five countries on the others represented as a percentage, and the import tariffs levied by each country on other countries are represented in brackets(in billion USD).

	US	France	India	Japan	UK
US		20% (6)	40% (4.5)	30% (7)	30% (3)
France	30% (5.5)		40% (6.5)	30% (4)	40% (4.5)
India	40% (9)	50% (5)		50% (3.5)	20% (5)
Japan	40% (4)	30% (3)	50% (8)		40% (3)
UK	20% (2.5)	30% (4)	30% (3)	40% (6)	

Japan's export to India would be the same as India's import from Japan.

India is charging a 50% tariff on Japan, and the tariff by India on Japan equals 3.5 billion USD.

So, 50% of the imports equals 3.5 billion USD.

$$\frac{50}{100} \times \text{Imports} = 3.5$$

$$\text{Imports} = 3.5 \times 2 = 7 \text{ billion USD.}$$

The value of imports by India from Japan = Japan's export to India = 7 billion USD.

Hence, the correct answer is option C.

[VIEW SOLUTION](#)



CAT Syllabus PDF



31. **B**

The values given in both charts together are represented in the table below, with the import tariff percentages charged by each of the five countries on the others represented as a percentage, and the import tariffs levied by each country on other countries are represented in brackets(in billion USD).

	US	France	India	Japan	UK
US		20% (6)	40% (4.5)	30% (7)	30% (3)
France	30% (5.5)		40% (6.5)	30% (4)	40% (4.5)
India	40% (9)	50% (5)		50% (3.5)	20% (5)
Japan	40% (4)	30% (3)	50% (8)		40% (3)
UK	20% (2.5)	30% (4)	30% (3)	40% (6)	

Option A) Exports by Japan to the UK would be the same as the UK's imports from Japan.

The UK is charging a 40% tariff on Japan, and the tariff imposed by the UK on Japan equals 6 billion USD.

So, 40% of the imports equals 6 billion USD.

$$\frac{40}{100} \times \text{Imports} = 6$$

$$\text{Imports} = 3 \times 5 = 15 \text{ billion USD.}$$

The value of imports by the UK from Japan = Japan's export to the UK = 15 billion USD.

Option B) Imports by the US from France.

The US is charging a 20% tariff on France, and the tariff by the US on France equals 6 billion USD.

So, 20% of the imports equals 6 billion USD.

$$\frac{20}{100} \times \text{Imports} = 6$$

$$\text{Imports} = 6 \times 5 = 30 \text{ billion USD.}$$

The value of imports by the US from France = 30 billion USD.

Option C) Exports by France to Japan would be the same as Japan's imports from France.

Japan is charging a 30% tariff on France, and the tariff by Japan on France equals 3 billion USD.

So, 30% of the imports equals 3 billion USD.

$$\frac{30}{100} \times \text{Imports} = 3$$

$$\text{Imports} = 1 \times 10 = 10 \text{ billion USD.}$$

The value of imports by Japan from France = France's export to Japan = 10 billion USD.

Option D) Imports by France from India.

France is charging a 40% tariff on India, and the tariff by France on India equals 6.5 billion USD.

So, 40% of the imports equals 6.5 billion USD.

$$\frac{40}{100} \times \text{Imports} = 6.5$$

$$\text{Imports} = 6.5 \times 2.5 = 16.25 \text{ billion USD.}$$

The value of imports by France from India = 16.25 billion USD.

Out of all the options, the value of imports by the US from France is the highest.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

32. D

The values given in both charts together are represented in the table below, with the import tariff percentages charged by each of the five countries on the others represented as a percentage, and the import tariffs levied by each country on other countries are represented in brackets(in billion USD).

	US	France	India	Japan	UK
US		20% (6)	40% (4.5)	30% (7)	30% (3)
France	30% (5.5)		40% (6.5)	30% (4)	40% (4.5)
India	40% (9)	50% (5)		50% (3.5)	20% (5)
Japan	40% (4)	30% (3)	50% (8)		40% (3)
UK	20% (2.5)	30% (4)	30% (3)	40% (6)	

For France - Trade surplus/trade deficit of France with the US = Exports from France to the US - Imports from the US to France

Exports by France to the US would be the same as the US's imports from France.

The US is charging a 20% tariff on India, and the tariff imposed by the US on France equals 6 billion USD.

So, 20% of the imports equals 6 billion USD.

$$\frac{20}{100} \times \text{Imports} = 6$$

$$\text{Imports} = 6 \times 5 = 30 \text{ billion USD.}$$

The value of imports by the US from France = France's export to the US = 30 billion USD.

Imports by France from the US can be calculated as,

France is charging a 30% tariff on the US, and the tariff imposed by France on the US equals 5.5 billion USD.

So, 30% of the imports equals 5.5 billion USD.

$$\frac{30}{100} \times \text{Imports} = 5.5$$

$$\text{Imports} = 5.5 \times 3.34 = 18.34 \text{ billion USD.}$$

The value of imports by France from the US = 18.34 billion USD.

We can clearly see that the Imports are less than the exports for France from the US. So, the trade surplus can be calculated as,

$$\text{Trade surplus} = \text{Exports from France to the US} - \text{Imports from France to the US} = 30 \text{ billion USD} - 18.34 \text{ billion USD} = 11.67 \text{ billion USD.}$$

So, there is a surplus of 11.67 billion USD.

For the UK - The Trade surplus/trade deficit of the UK with the US = Exports from the UK to the US - Imports from the US to the UK

Exports by the UK to the US would be the same as the US's imports from the UK.

The US is charging a 30% tariff on the UK, and the tariff imposed by the US on the UK equals 3 billion USD.

So, 30% of the imports equals 3 billion USD.

$$\frac{30}{100} \times \text{Imports} = 3$$

$$\text{Imports} = 1 \times 10 = 10 \text{ billion USD.}$$

The value of imports by the US from the UK = The UK's export to the US = 10 billion USD.

Imports by the UK from the US can be calculated as,

The UK is charging a 20% tariff on the US, and the tariff imposed by the UK on the US equals 2.5 billion USD.

So, 20% of the imports equals 2.5 billion USD.

$$\frac{20}{100} \times \text{Imports} = 2.5$$

$$\text{Imports} = 2.5 \times 5 = 12.5 \text{ billion USD.}$$

The value of imports by the UK from the US = 12.5 billion USD.

We can clearly see that the Imports are greater than the exports for the UK from the US. So, the trade deficit can be calculated as,

$$\text{Trade deficit} = \text{Exports from the UK to the US} - \text{Imports from the UK to the US} = 10 \text{ billion USD} - 12.5 \text{ billion USD} = -2.5 \text{ billion USD.}$$

So, there is a deficit of 2.5 billion USD.

We can see that France has a surplus and the UK has a deficit with the UK.

Hence, the correct answer is option D.

 [VIEW SOLUTION](#)

33. B

The values given in both charts together are represented in the table below, with the import tariff percentages charged by each of the five countries on the others represented as a percentage, and the import tariffs levied by each country on other countries are represented in brackets(in billion USD).

	US	France	India	Japan	UK
US		20% (6)	40% (4.5)	30% (7)	30% (3)
France	30% (5.5)		40% (6.5)	30% (4)	40% (4.5)
India	40% (9)	50% (5)		50% (3.5)	20% (5)
Japan	40% (4)	30% (3)	50% (8)		40% (3)
UK	20% (2.5)	30% (4)	30% (3)	40% (6)	

Trade surplus/trade deficit of India with the UK = Exports from India to the UK - Imports from India to the UK.
Exports by India to the UK would be the same as the UK's imports from India.

The UK is charging a 30% tariff on India, and the tariff imposed by the UK on India equals 3 billion USD.

So, 30% of the imports equals 3 billion USD.

$$\frac{30}{100} \times \text{Imports} = 3$$

$$\text{Imports} = 1 \times 10 = 10 \text{ billion USD.}$$

The value of imports by the UK from India = India's export to the UK = 10 billion USD.

Imports by India from the UK can be calculated as,

India is charging a 20% tariff on the UK, and the tariff imposed by India on the UK equals 5 billion USD.

So, 20% of the imports equals 5 billion USD.

$$\frac{20}{100} \times \text{Imports} = 5$$

$$\text{Imports} = 5 \times 5 = 25 \text{ billion USD.}$$

The value of imports by India from the UK = 25 billion USD.

We can clearly see that the Imports are greater than the exports for India from the UK. So, the trade deficit can be calculated as,

Trade deficit = Exports from India to the UK - Imports from India to the UK = 10 billion USD - 25 billion USD = -15 billion USD.

So, there is a deficit of 15 billion USD.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

34. D

It is given,

	Revenue			Cost incurred		
Company	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

Company A:

Revenue = 240 and cost incurred = 180

Profit = 240 - 180 = 60

Company B:

Revenue = 220 and cost incurred = 145

$$\text{Profit} = 220 - 145 = 75$$

Company C:

$$\text{Revenue} = 195 \text{ and cost incurred} = 110$$

$$\text{Profit} = 195 - 110 = 85$$

Company D:

$$\text{Revenue} = 140 \text{ and cost incurred} = 160$$

No profit.

Company C had the highest annual profit.

The answer is option D.

[VIDEO SOLUTION](#)

35. D

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

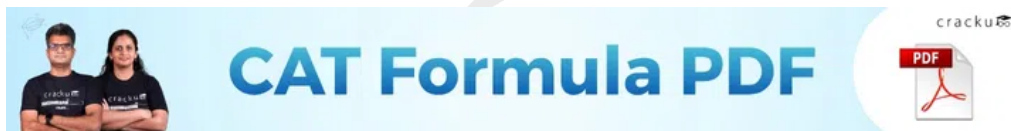
For all the companies in all three years, cost incurred is less than Revenue except for D in 2020.

Revenue is 20 and cost incurred is 50

Company D experienced the highest annual loss in 2020.

The answer is option D.

[VIDEO SOLUTION](#)



36. A

It is given,

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

$$\text{Measure of A's performance in 2019} = \frac{90-85}{85} = \frac{5}{85} = 0.06$$

$$\text{Measure of B's performance in 2019} = \frac{100-75}{75} = \frac{25}{75} = 0.33$$

$$\text{Measure of C's performance in 2019} = \frac{25-20}{25} = \frac{5}{25} = 0.2$$

$$\text{Measure of D's performance in 2019} = \frac{50-40}{40} = \frac{10}{40} = 0.25$$

Company A had the lowest value.

The answer is option A.

[VIDEO SOLUTION](#)

37. A

It is given,

Company	Strength			New hires		
	2019	2020	2021	2019	2020	2021
A	150	140	150	20	35	25
B	210	240	250	35	45	30
C	320	320	320	45	40	35
D	400	410	400	30	35	40

Company A:

The number of employees in the beginning of 2019 = 150

The number of employees hired in 2019 = 20

The number of employees should be at the beginning of 2020 is $150 + 20$, i.e. 170 but there are 140 only. This implies 30 left company A in 2019.

The number of employees in the beginning of 2020 = 140

The number of employees hired in 2020 = 35

The number of employees should be at the beginning of 2021 is $140 + 35$, i.e. 175 but there are 150 only. This implies 25 left company A in 2020.

The number of employees left company A in 2019 and 2020 = $30 + 25 = 55$

Company B:

Similarly, the number of employees left company B in 2019 = $210 + 35 - 240 = 5$

The number of employees left company B in 2020 = $240 + 45 - 250 = 35$

The number of employees left company B in 2019 and 2020 = $5 + 35 = 40$

Company C:

Similarly, the number of employees left company C in 2019 = $320 + 45 - 320 = 45$

The number of employees left company C in 2020 = $320 + 40 - 320 = 40$

The number of employees left company C in 2019 and 2020 = $45 + 40 = 85$

Company D:

Similarly, the number of employees left company D in 2019 = $400 + 30 - 410 = 20$

The number of employees left company D in 2020 = $410 + 35 - 400 = 45$

The number of employees left company D in 2019 and 2020 = $20 + 45 = 65$

The total number of employees lost in 2019 and 2020 is least for company B.

The answer is option A.

 VIDEO SOLUTION

38. C

It is given,

Company	Revenue			Cost incurred		
	2019	2020	2021	2019	2020	2021
A	90	90	60	85	65	30
B	100	90	30	75	40	30
C	25	70	100	20	60	30
D	50	20	70	40	50	70

	Strength			New hires		
Company	2019	2020	2021	2019	2020	2021
A	150	140	150	20	35	25
B	210	240	250	35	45	30
C	320	320	320	45	40	35
D	400	410	400	30	35	40

$$\text{Company A} = \frac{90-65}{145} = \frac{25}{145} = \frac{50}{290}$$

$$\text{Company B} = \frac{90-40}{245} = \frac{50}{245}$$

$$\text{Company C} = \frac{70-60}{320} = \frac{10}{320}$$

Company B had the highest profit per employee.

The answer is option C.

[VIDEO SOLUTION](#)

39. D

The Pat per employee for the firm in 2023 will be:

$$\text{Firm E: } \frac{3500}{1400} = \frac{35}{14} = \frac{5}{2}$$

$$\text{Firm A: } \frac{3900}{1300} = 3$$

$$\text{Firm F: } \frac{3200}{1000} = 3.2$$

$$\text{Firm C: } \frac{3000}{800} = \frac{15}{4} = 3.75$$

The value if maximum for firm C, therefore, Option D is the correct answer.

[VIDEO SOLUTION](#)

40. C

Since we are given that the annual growth rate should be taken as constant, we can simply compare the ratio changes in the firm's PAT in 2019 and 2023 and compare them.

$$\text{Firm A: } 3000 \text{ to } 3900; \text{ an increase of } 900, \text{ giving the growth rate to be } \frac{900}{3000} = \frac{3}{10} = 0.3$$

$$\text{Firm C: } 2400 \text{ to } 3000; \text{ an increase of } 600, \text{ giving the growth rate to be } \frac{600}{2400} = \frac{1}{4} = 0.25$$

$$\text{Firm E: } 2400 \text{ to } 3500; \text{ an increase of } 1100, \text{ giving the growth rate to be } \frac{1100}{2400} = \frac{11}{24} \approx 0.45$$

$$\text{Firm B: } 2800 \text{ to } 3800; \text{ an increase of } 1000, \text{ giving the growth to be } \frac{1000}{2800} = \frac{5}{14} \approx 0.35$$

We can see that of the given ratios, the growth rate is highest for firm E (closest to 0.5). Therefore, the highest ARG, among the given options, will be for firm E.

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)



41. D

This can be a little difficult to solve since the bubbles are not closed within a fixed boundary, but in such instances, we should approximate the most visible cases.

Like the bubble of C in 2019 is almost 3 units vertically long in diameter, while that in 2023 is 2 units long.

The ratio of the amount spend on R&D on 2019 to 2024 would be given by: $\frac{2400\pi(1.5)^2}{3000\pi(1)^2} = \frac{8}{10} \times \frac{9}{4} = 9 : 5$

Which is option D.

 VIDEO SOLUTION

42. **B**

We calculated the PAT per employee for F, C and E in the last question, using that information and calculating the PAT per employee for firm D to be $\frac{2400}{800} = 3$

For firms F, C and D, we can see that the area is the same hence the only factor to consider is PAT per employee F is 3.2, C is 3.75, and D is 3

This leaves D as the minimum.

Comparing E and D (taking the proportionality constant as k)

E has PAT per employee as 2.5 with a radius of 1.5 units, the PRD value per employee would be $\frac{5}{2}k\pi \left(\frac{3}{2}\right)^2 = k\pi \times \frac{5 \times 9}{8}$

D has PAT per employee as 3, with a radius of 1 unit. the PRD value per employee would be $3k\pi (1)^2 = 3k\pi$

We can see that the value would be greater for E, therefore firm D will have the lowest R&D spending per employee in 2023

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

43. **6**

Comparing the 1st and 2nd graphs, we should be able to determine the type of puzzle.

From the second graph, we can say that there are 4 visual-based puzzles in total, which makes the number-based puzzles 6.

Let us follow the green line in both graphs:

The 1st visual-based puzzle was completed by Green in 2 minutes from the 2nd graph. From the 1st graph at the 2nd minute, the puzzle that was solved by Green was 1. So, we can say that 1 is a visual-based puzzle.

The 2nd visual-based puzzle was completed by Green in 12 minutes from the 2nd graph. From the 1st graph at the 12th minute, the puzzle that was solved by Green was 4. So, we can say that 4 is a visual-based puzzle.

The 3rd visual-based puzzle was completed by Green in 25 minutes from the 2nd graph. From the 1st graph at the 25th minute, the puzzle that was solved by Green was 8. So, we can say that 8 is a visual-based puzzle.

The 4th visual-based puzzle was completed by Green in 29 minutes from the 2nd graph. From the 1st graph at the 29th minute, the puzzle that was solved by Green was 9. So, we can say that 9 is a visual-based puzzle.

We can conclude that puzzles 1, 4, 8, and 9 are visual-based, while puzzles 2, 3, 5, 6, 7, and 10 are number-based.

The fourth number-based puzzle is 6th puzzle in the order.

Hence, the correct answer is 6.

 VIEW SOLUTION

44. **B**

The left plot shows, for each competitor, a step (staircase) curve whose height at any time t equals the number of puzzles that competitor has completed by minute t.

To find who solved the most puzzles by the 20th minute, draw a vertical line at t=20 (or simply read the height of each competitor's colored stair at the vertical line x=20).

Reading the four curves at $t=20$:

The green staircase (Koushik) is at the highest horizontal level at $t=20$.

The yellow (Chandranath) and red (Anirbid) staircases are lower than the green ones at that time.

The blue (Suranjan) staircase is the lowest of the four at $t=20$.

Since Koushik's value is greater than every other competitor's at $t=20$, Koushik has solved the largest number of puzzles by the 20th minute.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

45. D

Comparing the 1st and 2nd graphs, we should be able to determine the type of puzzle.

From the second graph, we can say that there are 4 visual-based puzzles in total, which makes the number-based puzzles 6.

Let us follow the green line in both graphs:

The 1st visual-based puzzle was completed by Green in 2 minutes from the 2nd graph. From the 1st graph at the 2nd minute, the puzzle that was solved by Green was 1. So, we can say that 1 is a visual-based puzzle.

The 2nd visual-based puzzle was completed by Green in 12 minutes from the 2nd graph. From the 1st graph at the 12th minute, the puzzle that was solved by Green was 4. So, we can say that 4 is a visual-based puzzle.

The 3rd visual-based puzzle was completed by Green in 25 minutes from the 2nd graph. From the 1st graph at the 25th minute, the puzzle that was solved by Green was 8. So, we can say that 8 is a visual-based puzzle.

The 4th visual-based puzzle was completed by Green in 29 minutes from the 2nd graph. From the 1st graph at the 29th minute, the puzzle that was solved by Green was 9. So, we can say that 9 is a visual-based puzzle.

We can conclude that puzzles 1, 4, 8, and 9 are visual-based, while puzzles 2, 3, 5, 6, 7, and 10 are number-based.

The average time taken by Anirbid to solve the number-based puzzles can be calculated as,

Time for 1st number-based puzzle(2nd puzzle) from 1st graph = $9 - 4 = 5$ minutes.

Time for 2nd number-based puzzle(3rd puzzle) from 1st graph = $11 - 9 = 2$ minutes.

Time for 3rd number-based puzzle(5th puzzle) from 1st graph = $19 - 14 = 5$ minutes.

Time for 4th number-based puzzle(6th puzzle) from 1st graph = $22 - 19 = 3$ minutes.

Time for 5th number-based puzzle(7th puzzle) from 1st graph = $27 - 22 = 5$ minutes.

Time for 6th number-based puzzle(10th puzzle) from 1st graph = $37 - 33 = 4$ minutes.

$$\text{Average} = \frac{5 + 2 + 5 + 3 + 5 + 4}{6} = \frac{24}{6} = 4 \text{ minutes}$$

The average time taken by Anirbid to solve the number-based puzzles is 4 minutes.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

CAT Percentile Predictor



46. 2

Comparing the 1st and 2nd graphs, we should be able to determine the type of puzzle.

From the second graph, we can say that there are 4 visual-based puzzles in total, which makes the number-based puzzles 6.

Let us follow the green line in both graphs:

The 1st visual-based puzzle was completed by Green in 2 minutes from the 2nd graph. From the 1st graph at the 2nd minute, the puzzle that was solved by Green was 1. So, we can say that 1 is a visual-based puzzle.

The 2nd visual-based puzzle was completed by Green in 12 minutes from the 2nd graph. From the 1st graph at the 12th minute, the puzzle that was solved by Green was 4. So, we can say that 4 is a visual-based puzzle.

The 3rd visual-based puzzle was completed by Green in 25 minutes from the 2nd graph. From the 1st graph at the 25th minute, the puzzle that was solved by Green was 8. So, we can say that 8 is a visual-based puzzle.

The 4th visual-based puzzle was completed by Green in 29 minutes from the 2nd graph. From the 1st graph at the 29th minute, the puzzle that was solved by Green was 9. So, we can say that 9 is a visual-based puzzle.

We can conclude that puzzles 1, 4, 8, and 9 are visual-based, while puzzles 2, 3, 5, 6, 7, and 10 are number-based.

So, the third visual puzzle is 8, and the time taken by Suranjan(blue) to complete the 3rd visual-based puzzle, which is the 8th puzzle, can be calculated as,

The time taken by Suranjan to solve the 8th puzzle from graph 1 = $28 - 26 = 2$ minutes.

Hence, the correct answer is 2.

[VIEW SOLUTION](#)

47. A

All the states which satisfy the condition for 'Low monsoon state' are Gujrat (+25%), Karnataka (+20%), Rajasthan (+15%), MP (+10%), Assam (-10%), WB (-30%), Jharkhand (-35%), Delhi (-40%) and Manipur (-60%).

The median of all the deviation is -10% Assam.

[VIDEO SOLUTION](#)

48. D

The states Assam, WB, Jharkhand, Delhi and Manipur satisfy the conditions given in the question.

The actual rainfall of all these states in 2019 are 600,600,400,300,400

Average of these states = $2300/5 = 460$ mm

[VIDEO SOLUTION](#)

49. A

The states which satisfy the condition given in the question:

Maharashtra, Mizoram, Sikkim, Goa, Arunachal, Kerla, Meghalaya.....7 states

The 'Heavy Monsoon States' have a negative deviation: Arunachal, Kerla, Meghalaya

= $3/7 \times 100 = 42.86\%$

Option A

[VIDEO SOLUTION](#)

50. C

The actual rainfall in Gujarat in 2019 is 20% more than the rainfall in 2009.

So If the actual rainfall in 2009 = x mm

Then the actual rainfall in 2019 = $1.2x$ mm

Actual rainfall in 2019 = 600mm

Then, actual rainfall in 2009 = 500mm

As deviation is +25% so average 2009 - 2018 is $600/1.25 = 480$

LPA 2019 = $(480 \times 10 - 500 + 600)/10 = 490\text{mm}$

Answer C

[VIDEO SOLUTION](#)



51. D

We can make the following table from the web chart given in the question:

Revenue				Cost			
	Clothing	Produce	Electronics		Clothing	Produce	Electronics
2016	20	30	50		30	40	30
2017	30	40	30		30	30	40
2018	40	40	20		20	50	30

If we consider the total cost in the year

2016 as 100, then according to Statement 1, the total revenue in 2016 must be 200.

The store's revenue doubled from 2016 to 2017, thus the total revenue in the year 2017 = 400.

We can find the revenue for the individual department in the year 2017, from the table.

There was no profit from the Electronics department in 2017, thus, we can find the total cost in 2017 = 300

Considering the statement 4, we can find the total revenue in 2018 and tabulate the following table.

Cost							
	Clothing		Produce		Electronics		Total
2016	30%	30	40%	40	30%	30	100
2017	30%	90	30%	90	40%	120	300
2018	20%	40	50%	100	30%	60	200

Revenue							
	Clothing		Produce		Electronics		
2016	20%	40	30%	60	50%	100	200
2017	30%	120	40%	160	30%	120	400
2018	40%	100	40%	100	20%	50	250

The ratio of revenue generated from the Produce department in 2017 to that in 2018 = $160:100 = 8:5$

[VIDEO SOLUTION](#)

52. 70

We can make the following table from the web chart given in the question:

Revenue				Cost			
	Clothing	Produce	Electronics		Clothing	Produce	Electronics
2016	20	30	50		30	40	30
2017	30	40	30		30	30	40
2018	40	40	20		20	50	30

If we consider the total cost in the year

2016 as 100, then according to Statement 1, the total revenue in 2016 must be 200.

The store's revenue doubled from 2016 to 2017, thus the total revenue in the year 2017 = 400.

We can find the revenue for the individual department in the year 2017, from the table.

There was no profit from the Electronics department in 2017, thus, we can find the total cost in 2017= 300

Considering the statement 4, we can find the total revenue in 2018 and tabulate the following table.

Cost							
	Clothing		Produce		Electronics		Total
2016	30%	30	40%	40	30%	30	100
2017	30%	90	30%	90	40%	120	300
2018	20%	40	50%	100	30%	60	200

Revenue							
	Clothing		Produce		Electronics		
2016	20%	40	30%	60	50%	100	200
2017	30%	120	40%	160	30%	120	400
2018	40%	100	40%	100	20%	50	250

Profit in 2016 = $200 - 100 = 100$

Profit in the electronic department in 2016= $100 - 30 = 70$

The total profits= for the store in 2016 were from the Electronics department 70%

 VIDEO SOLUTION

53. C

We can make the following table from the web chart given in the question:

Revenue				Cost			
	Clothing	Produce	Electronics		Clothing	Produce	Electronics
2016	20	30	50		30	40	30
2017	30	40	30		30	30	40
2018	40	40	20		20	50	30

If we consider the total cost in the year

2016 as 100, then according to Statement 1, the total revenue in 2016 must be 200.

The store's revenue doubled from 2016 to 2017, thus the total revenue in the year 2017 = 400.

We can find the revenue for the individual department in the year 2017, from the table.

There was no profit from the Electronics department in 2017, thus, we can find the total cost in 2017= 300

Considering the statement 4, we can find the total revenue in 2018 and tabulate the following table.

Cost							
	Clothing		Produce		Electronics		Total
2016	30%	30	40%	40	30%	30	100
2017	30%	90	30%	90	40%	120	300
2018	20%	40	50%	100	30%	60	200

Revenue							
	Clothing		Produce		Electronics		
2016	20%	40	30%	60	50%	100	200
2017	30%	120	40%	160	30%	120	400
2018	40%	100	40%	100	20%	50	250

Profit percentage in 2017 = $(400-300)/300\% = 33.33\%$

Profit percentage in 2018 = $(250-200)/200\% = 25\%$

The approximate difference in profit percentages of the store in 2017 and 2018 = $(33.33-25)\% = 8.33\%$

Option C.

 VIDEO SOLUTION

54.25

We can make the following table from the web chart given in the question:

Revenue				Cost			
	Clothing	Produce	Electronics		Clothing	Produce	Electronics
2016	20	30	50		30	40	30
2017	30	40	30		30	30	40
2018	40	40	20		20	50	30

If we consider the total cost in the year

2016 as 100, then according to Statement 1, the total revenue in 2016 must be 200.

The store's revenue doubled from 2016 to 2017, thus the total revenue in the year 2017 = 400.

We can find the revenue for the individual department in the year 2017, from the table.

There was no profit from the Electronics department in 2017, thus, we can find the total cost in 2017 = 300

Considering the statement 4, we can find the total revenue in 2018 and tabulate the following table.

Cost							
	Clothing		Produce		Electronics		Total
2016	30%	30	40%	40	30%	30	100
2017	30%	90	30%	90	40%	120	300
2018	20%	40	50%	100	30%	60	200

Revenue							
	Clothing		Produce		Electronics		
2016	20%	40	30%	60	50%	100	200
2017	30%	120	40%	160	30%	120	400
2018	40%	100	40%	100	20%	50	250

The percentage profit of the store in 2018= $(250-200)/200 = 25\%$

 VIDEO SOLUTION

